

METHODS OF DIFFERENTIATION

JEE EXERCISES

Exercise J-I

BASED ON DIFFERENTIATION TECHNIQUES

1. If $y = \log(x + \sqrt{x^2 + a^2})$ then dy/dx equals :

यदि $y = \log(x + \sqrt{x^2 + a^2})$ तो dy/dx बराबर है :

(A) $\frac{x}{\sqrt{x^2 + a^2}}$

(B) $\frac{1}{\sqrt{x^2 + a^2}}$

(C) $\frac{-1}{\sqrt{x^2 + a^2}}$

(D) $\frac{1}{\sqrt{x^2 - a^2}}$

Ans. B

Sol. $y = \log(x + \sqrt{x^2 + a^2})$

$$\frac{dy}{dx} = \frac{1}{x + \sqrt{x^2 + a^2}} \times \left(1 + \frac{2x}{2\sqrt{x^2 + a^2}} \right)$$

$$\Rightarrow \frac{1}{\sqrt{x^2 + a^2}}$$

2. If $y = 1 + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ then dy/dx equals :

यदि $y = 1 + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ तो dy/dx बराबर है :

(A) $y - x - 1$

(B) $y + x - 1$

(C) $y + x$

(D) $y - x$

Ans. B

Sol. $y = 1 + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

$$\frac{dy}{dx} = \frac{2x}{2!} + \frac{3x^2}{3!} + \frac{4x^3}{4!} + \dots$$

$$\frac{dy}{dx} = \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\frac{dy}{dx} = 1 - 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\frac{dy}{dx} = y + x - 1$$

3. The derivative of $\tan x^\circ$ is :

$\tan x^\circ$ का अवकलज है :

(A) $\left(\frac{\pi}{180}\right) \cos^2 x$ (B) $\left(\frac{\pi}{180}\right) \sec^2 x$ (C) $\left(\frac{\pi}{180}\right) \cos^2 x^\circ$ (D) $\left(\frac{\pi}{180}\right) \sec^2 x^\circ$

Ans. D

Sol. $y = \tan x^\circ$

$$= \tan\left(\frac{\pi x}{180}\right)$$

$$= \frac{\pi}{180} \sec^2 x^\circ$$

4. If $x^y + y^x = a^b$ then $\frac{dy}{dx}$ equals :

(A) $\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$ (B) $-\frac{x^y \log x + x.y^{x-1}}{y.x^{y-1} + \log y y^x}$ (C) $-\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$ (D) None of these

यदि $x^y + y^x = a^b$ then $\frac{dy}{dx}$ बराबर है :

(A) $\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$ (B) $-\frac{x^y \log x + x.y^{x-1}}{y.x^{y-1} + \log y y^x}$ (C) $-\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$ (D) इनमें से कोई नहीं

Ans. C

Sol. $x^y + y^x = a^b$

$$x^y \ln x y' + yx^{y-1} + xy^{x-1}y' + y^x \ln y = 0$$

$$y' (x^y \ln x + xy^{x-1}) = -(yx^{y-1} + y^x \ln y)$$

$$\frac{dy}{dx} = \frac{-(yx^{y-1} + y^x \ln y)}{x^y \ln x + xy^{x-1}}$$

5. If $\sin y = x \sin(a+y)$ then dy/dx is :

[AIEEE-2002]

(A) $\frac{\sin(a+y)}{\sin^2 a}$ (B) $\frac{\sin^2(a+y)}{\sin a}$ (C) $\frac{\sin(a+y)}{\sin^2 a}$ (D) None of these

यदि $\sin y = x \sin(a+y)$ तो dy/dx होगा :

[AIEEE-2002]

- (A) $\frac{\sin(a+y)}{\sin^2 a}$ (B) $\frac{\sin^2(a+y)}{\sin a}$ (C) $\frac{\sin(a+y)}{\sin^2 a}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $x = \frac{\sin y}{\sin(a+y)}$

$$\frac{dx}{dy} = \frac{\cos y \cdot \sin(a+y) - \sin y \cos(a+y)}{\sin^2(a+y)}$$

$$\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$$

6. If $y = (1+x^{1/4})(1+x^{1/2})(1-x^{1/4})$, then dy/dx equals :

यदि $y = (1+x^{1/4})(1+x^{1/2})(1-x^{1/4})$, तो dy/dx बराबर होगा :

- (A) -1 (B) 1 (C) x (D) $\sqrt{x}$

Ans. A

Sol. $y = (1+x^{1/4})(1+x^{1/2})(1-x^{1/4})$

$$y = \left(1+x^{\frac{1}{2}}\right) \left(1+x^{\frac{1}{4}}\right) \left(1-x^{\frac{1}{4}}\right)$$

$$y = \left(1+x^{\frac{1}{2}}\right) \left(1-x^{\frac{1}{2}}\right)$$

$$y = 1 - x$$

$$\frac{dy}{dx} = -1$$

7. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, then $\frac{dy}{dx}$ equals :

- (A) $\frac{1}{(1+x)^2}$ (B) $-\frac{1}{(1+x)^2}$ (C) $\frac{1}{1+x^2}$ (D) None of these

यदि $x\sqrt{1+y} + y\sqrt{1+x} = 0$, तो $\frac{dy}{dx}$ बराबर है :

(A) $\frac{1}{(1+x)^2}$

(B) $-\frac{1}{(1+x)^2}$

(C) $\frac{1}{1+x^2}$

(D) इनमें से कोई नहीं

Ans. B

Sol. $x^2(1+y) = y^2(1+x) \Rightarrow x^2 - y^2 + xy(x-y) = 0$

$\Rightarrow (x-y)(x+y+xy) = 0$

Now $x \neq y$ [does not satisfy the given equation]

$\therefore x + y + xy = 0 \Rightarrow y = \frac{-x}{1+x}$

$\therefore \frac{dy}{dx} = \frac{-(1+x) + x}{(1+x)^2} = -\frac{1}{(1+x)^2}$

8. If $y = \sin^{-1} \sqrt{\sin x}$, then $\frac{dy}{dx}$ equals :यदि $y = \sin^{-1} \sqrt{\sin x}$, then $\frac{dy}{dx}$ बराबर है :

(A) $\frac{2\sqrt{\sin x}}{\sqrt{1+\sin x}}$

(B) $\frac{\sqrt{\sin x}}{\sqrt{1-\sin x}}$

(C) $\frac{1}{2} \sqrt{1 + \operatorname{cosec} x}$

(D) $\frac{1}{2} \sqrt{1 - \operatorname{cosec} x}$

Ans. C

Sol. $y = \sin^{-1} \sqrt{\sin x}$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-\sin x}} \times \frac{\cos x}{2\sqrt{\sin x}}$$

$$= \frac{1}{\sqrt{\frac{(1+\sin x)}{1+\sin x}} \times (1-\sin x)} \times \frac{\cos x}{2\sqrt{\sin x}}$$

$$= \frac{\sqrt{1+\sin x}}{\sqrt{1-\sin^2 x}} \times \frac{\cos x}{2\sqrt{\sin x}}$$

$$= \frac{\sqrt{1+\sin x}}{\cos x} \times \frac{\cos x}{2\sqrt{\sin x}}$$

$$= \frac{1}{2} \left(\sqrt{1 + \operatorname{cosec} x} \right)$$

9. If $x^y y^x = 1$, then $\frac{dy}{dx}$ equals :

यदि $x^y y^x = 1$, तो $\frac{dy}{dx}$ बराबर है :

(A) $\frac{x(y+x \log y)}{y(x+y \log x)}$ (B) $-\frac{x(x+y \log y)}{y(y+x \log x)}$ (C) $\frac{y(y+x \log y)}{x(x+y \log x)}$ (D) $-\frac{y(y+x \log y)}{x(x+y \log x)}$

Ans. D

Sol. $x^y y^x = 1$

$$\log x^y + \log y^x = \log 1$$

$$y \log x + x \log y = 0$$

$$\frac{y}{x} + \frac{dy}{dx} \log x + \log y + \frac{x}{y} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} \left(\log x + \frac{x}{y} \right) = - \left(\frac{y}{x} + \log y \right)$$

$$\frac{dy}{dx} = -\frac{y}{x} \left(\frac{y+x \log y}{x+y \log x} \right)$$

10. If $y = \tan^{-1} \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$, then $\frac{dy}{dx}$ equals :

यदि $y = \tan^{-1} \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$, तो $\frac{dy}{dx}$ बराबर है :

(A) $-\frac{1}{2\sqrt{1-x^2}}$ (B) $-\frac{1}{\sqrt{1-x^4}}$ (C) $-\frac{x}{\sqrt{1-x^4}}$ (D) $-\frac{x}{2\sqrt{1-x^4}}$

Ans. C

Sol. $y = \tan^{-1} \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$

$$= \tan^{-1} \left(\frac{1+x^2+1-x^2+2\sqrt{1-x^4}}{1+x^2-1-x^2} \right)$$

$$= \tan^{-1} \left(\frac{1 + \sqrt{1 - x^4}}{x^2} \right)$$

$$x^2 = \sin 2\theta$$

$$= \tan^{-1} \left(\frac{1 + \cos 2\theta}{\sin 2\theta} \right)$$

$$= \tan^{-1} (\cot \theta)$$

$$= \tan^{-1} \left(\tan \left(\frac{\pi}{2} - \theta \right) \right)$$

$$y = \frac{\pi}{2} - \theta$$

$$y = \frac{\pi}{2} - \frac{1}{2} \sin^{-1}(x^2)$$

$$\frac{dy}{dx} = \frac{-1}{2} \frac{1}{\sqrt{1 - x^4}} - 2x$$

$$= \frac{-x}{\sqrt{1 - x^4}}$$

11. If $f(x) = \log_x (\ln x)$, then at $x = e$, $f(x)$ equals :

यदि $f(x) = \log_x (\ln x)$, तो $x = e$ पर, $f(x)$ बराबर है :

(A) 0 (B) 1 (C) e (D) $1/e$

Ans. D

Sol. $f(x) = \log_x (\ln x)$

$$\Rightarrow f(x) = \frac{\ln(\ln x)}{\ln x}$$

$$\Rightarrow f'(x) = \frac{\ln x \left(\frac{1}{\ln x} \cdot \frac{1}{x} \right) - \frac{1}{x} \ln(\ln x)}{(\ln x)^2}$$

$$\Rightarrow f'(e) = \frac{\frac{1}{e} - 0}{1} = 1/e$$

12. If $y = e^{3x} \sin 4x$, then the value of dy/dx is :

यदि $y = e^{3x} \sin 4x$, तो dy/dx का मान है :

(A) $e^{3x} \sin \left(4x + \tan^{-1} \frac{4}{3} \right)$

(B) $e^{3x} \cos \left(4x + \tan^{-1} \frac{4}{3} \right)$

(C) $5e^{3x} \sin \left(4x + \tan^{-1} \frac{4}{3} \right)$

(D) $5e^{3x} \cos \left(4x + \tan^{-1} \frac{4}{3} \right)$

Ans. C

Sol. $y = e^{3x} \sin 4x$

$$\frac{dy}{dx} = 3e^{3x} \sin 4x + 4e^{3x} \cos 4x$$

$$= e^{3x} (3 \sin 4x + 4 \cos 4x)$$

$$= 5e^{3x} \left(\frac{3}{5} \sin 4x + \frac{4}{5} \cos 4x \right)$$

$$= 5e^{3x} \sin \left(4x + \tan^{-1} \frac{4}{3} \right)$$

13. If $x + y = x^y$, then dy/dx equals :

(A) $\frac{yx^{y-1} - 1}{1 - x^y \log x}$

(B) $\frac{y x^{y-1} - 1}{x^y \log x - 1}$

(C) $\frac{y x^{y-1} + 1}{x^y \log x + 1}$

(D) None of these

यदि $x + y = x^y$, तो dy/dx बराबर है :

(A) $\frac{yx^{y-1} - 1}{1 - x^y \log x}$

(B) $\frac{y x^{y-1} - 1}{x^y \log x - 1}$

(C) $\frac{y x^{y-1} + 1}{x^y \log x + 1}$

(D) इनमें से कोई नहीं

Ans. A

Sol. $x + y = x^y$

$$1 + \frac{dy}{dx} = yx^{y-1} + x^y \ln x \frac{dy}{dx}$$

$$\frac{dy}{dx} (1 - x^y \ln x) = yx^{y-1} - 1$$

$$\frac{dy}{dx} = \frac{yx^{y-1} - 1}{1 - x^y \ln x}$$

14. The derivative of x^{a^x} is :

(A) $x^{a^x} [a^x x^{-1} + a^x \log a \log x]$

(B) $x^{a^x} [xa^x + a^x \log x]$

(C) $x^{a^x} [a^x + xa^x \log x]$

(D) None of these

x^{a^x} का अवकलज है :

(A) $x^{a^x} [a^x x^{-1} + a^x \log a \log x]$

(B) $x^{a^x} [xa^x + a^x \log x]$

(C) $x^{a^x} [a^x + xa^x \log x]$

(D) None of these

Ans. A

Sol. $y = x^{a^x}$

$$\ln y = a^x \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = a^x \ln a \ln x + \frac{a^x}{x}$$

$$\frac{dy}{dx} = (x)^{a^x} \left[a^x \ln a \ln x + x^{-1} a^x \right]$$

15. If $(\cos x)^y = (\sin y)^x$, then dy/dx equals :

(A) $\frac{\log \sin y - y \tan x}{\log \cos x + x \cot y}$

(B) $\frac{\log \sin y + y \tan x}{\log \cos x - x \cot y}$

(C) $\frac{\log \sin y + y \tan x}{\log \cos x + x \cot y}$

(D) None of these

यदि $(\cos x)^y = (\sin y)^x$, तो dy/dx बराबर है :

(A) $\frac{\log \sin y - y \tan x}{\log \cos x + x \cot y}$

(B) $\frac{\log \sin y + y \tan x}{\log \cos x - x \cot y}$

(C) $\frac{\log \sin y + y \tan x}{\log \cos x + x \cot y}$

(D) इनमें से कोई नहीं

Ans. B

Sol. $(\cos x)^y = (\sin y)^x$

$$y \log(\cos x) = x \log(\sin y)$$

$$\frac{dy}{dx} \log(\cos x) + y \times \frac{-\sin x}{\cos x} = \log(\sin y) + \frac{x \cos y}{\sin y} \frac{dy}{dx}$$

$$\frac{dy}{dx} (\log(\cos x) - x \cot y) = y \tan + \log(\sin y)$$

$$\frac{dy}{dx} = \frac{y \tan x + \log(\sin y)}{\log(\cos x) - x \cot y}$$

16. If $y = \sin^{-1} [x \sqrt{1-x} - \sqrt{x} \sqrt{1-x^2}]$, then dy/dx can be equal to :

(A) $\frac{2}{1+x^2}$

(B) $\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x} \sqrt{1-x}}$

(D) None of these

यदि $y = \sin^{-1} [x \sqrt{1-x} - \sqrt{x} \sqrt{1-x^2}]$, तो dy/dx हो सकता है :

(A) $\frac{2}{1+x^2}$

(B) $\frac{1}{\sqrt{1-x^2}}$

(C) $\frac{1}{\sqrt{1-x^2}} - \frac{1}{2\sqrt{x} \sqrt{1-x}}$

(D) इनमें से कोई नहीं

Ans. C

Sol. $y = \sin^{-1} [x \sqrt{1-x} - \sqrt{x} \sqrt{1-x^2}]$

$$\therefore \sin^{-1} a \pm \sin^{-1} b = \sin^{-1} (a\sqrt{1-b^2} \pm b\sqrt{1-a^2})$$

$$y = \sin^{-1} x - \sin^{-1} \sqrt{x}$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{\sqrt{1-x}} \times \frac{1}{2\sqrt{x}}$$

17. The differential coefficient ($x > 0$) of $\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$ w.r.t $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$ can be equal to :

$\sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$ का $\cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$ के सापेक्ष अवकलन गुणांक ($x > 0$) हो सकता है :

(A) $1/2$

(B) $-1/2$

(C) 1

(D) -1

Ans. A

Sol. $y = \sin^{-1} \left(\frac{x}{\sqrt{1+x^2}} \right)$

$$x = \tan \theta$$

$$y = \sin^{-1} \left(\frac{\tan \theta}{\sec \theta} \right)$$

$$y = \sin^{-1}(\sin \theta)$$

$$y = \theta$$

$$y = \tan^{-1} x$$

$$\frac{dy}{dx} = \frac{1}{1+x^2}$$

$$\frac{dy}{dz} = \frac{1}{2}$$

$$z = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

$$x = \tan \phi$$

$$z = \cos^{-1} \left(\frac{1-\tan^2 \phi}{1+\tan^2 \phi} \right)$$

$$z = \cos^{-1}(\cos 2\phi)$$

$$z = 2\phi$$

$$z = 2 \tan^{-1} x$$

$$\frac{dz}{dx} = \frac{2}{1+x^2}$$

18. If $y = \sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\dots \infty}}}}$, then the value of dy/dx is :

यदि $y = \sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\dots \infty}}}}$, तो dy/dx का मान होगा :

(A) $\frac{xy^2}{2-y \log x}$ (B) $\frac{x^2}{y(2-y \log x)}$ (C) $\frac{y^2}{x(2-y \log x)}$ (D) $\frac{y^2}{x(2+y \log x)}$

Ans. C

Sol. $y = (\sqrt{x})^y$

$$\ln y = \frac{y}{2} \ln x$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{\ln x}{2} \frac{dy}{dx} + \frac{y}{2x}$$

$$\frac{dy}{dx} \left[\frac{1}{y} - \frac{\ln x}{2} \right] = \frac{y}{2x}$$

$$\frac{dy}{dx} \left[\frac{2 - y \ln x}{2y} \right] = \frac{y}{2x}$$

$$\frac{dy}{dx} = \frac{y^2}{x(2 - y \ln x)}$$

19. If $y = (\tan x)^{\tan x \dots \infty}$, then $\tan x \left(\frac{dy}{dx} \right)$ equals :

- (A) $\frac{y^2 \sec^2 x}{1 - y \log \tan x}$ (B) $\frac{y^2 \sec^2 x}{1 + y \log \tan x}$ (C) $\frac{y \sec^2 x}{1 - y \log \tan x}$ (D) None of these

यदि $y = (\tan x)^{\tan x \dots \infty}$, तो $\tan x \left(\frac{dy}{dx} \right)$ बराबर है :

- (A) $\frac{y^2 \sec^2 x}{1 - y \log \tan x}$ (B) $\frac{y^2 \sec^2 x}{1 + y \log \tan x}$ (C) $\frac{y \sec^2 x}{1 - y \log \tan x}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $y = (\tan x)^y$

$$\log y = y \log (\tan x)$$

$$\frac{1}{y} \frac{dy}{dx} = \frac{dy}{dx} \log(\tan x) + \frac{y}{\tan x} \sec^2 x$$

$$\frac{dy}{dx} \left(\frac{1}{y} - \log(\tan x) \right) = \frac{y \sec^2 x}{\tan x}$$

$$\frac{dy}{dx} = \frac{y^2 \sec^2 x}{\tan x (1 - y \log \tan x)}$$

$$\tan x \frac{dy}{dx} = \frac{y^2 \sec^2 x}{(1 - y \log \tan x)}$$

20. If $y = \frac{x}{a + \frac{x}{b + \frac{x}{a + \frac{x}{b + \dots}}}}$, then dy/dx equals :

- (A) $\frac{b}{a(b+2y)}$ (B) $\frac{b}{b+2y}$ (C) $\frac{a}{b(b+2y)}$ (D) None of these

यदि $y = \frac{x}{a + \frac{x}{b + \frac{x}{a + \frac{x}{b + \dots}}}}$, तो dy/dx बराबर होगा :

- (A) $\frac{b}{a(b+2y)}$ (B) $\frac{b}{b+2y}$ (C) $\frac{a}{b(b+2y)}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $y = \frac{x}{a + \frac{x}{b+y}} \Rightarrow y = \frac{x(b+y)}{ab+ay+x}$

$$\Rightarrow aby + ay^2 + xy = xb + xy$$

$$\Rightarrow ab \frac{dy}{dx} + 2ay \frac{dy}{dx} = b \Rightarrow \frac{dy}{dx} = \frac{b}{ab+2ay}$$

21. If $y = \log_y x$, then $\frac{dy}{dx} =$

यदि $y = \log_y x$, तो $\frac{dy}{dx} =$

- (A) $\frac{1}{x + \log y}$ (B) $\frac{1}{\log x (1 + y)}$ (C) $\frac{1}{x (1 + \log y)}$ (D) $\frac{1}{y + \log x}$

Ans. C

Sol. $y = \log_y x \quad y = \frac{\log x}{\log y}$

$$y \times \log y = \log x$$

$$\frac{dy}{dx} \log y + y \times \frac{1}{y} \frac{dy}{dx} = \frac{1}{x}$$

$$\frac{dy}{dx} (\log y + 1) = \frac{1}{x} \quad \boxed{\frac{dy}{dx} = \frac{1}{x(1 + \log y)}}$$

22. If $x = 3 \cos \theta - 2 \cos^3 \theta$ and $y = 3 \sin \theta - 2 \sin^3 \theta$, then $\frac{dy}{dx} =$

यदि $x = 3 \cos \theta - 2 \cos^3 \theta$ तथा $y = 3 \sin \theta - 2 \sin^3 \theta$, तो $\frac{dy}{dx} =$

- (A) $\cot \theta$ (B) $\sec \theta$ (C) $\tan \theta$ (D) $\csc \theta$

Ans. D

Sol. $x = 3 \cos \theta - 2 \cos^3 \theta$ $y = 3 \sin \theta - 2 \sin^3 \theta$

$$\frac{dx}{d\theta} = -3 \sin \theta + 6 \cos^2 \theta \times \sin \theta$$

$$\frac{dy}{d\theta} = 3 \cos \theta - 6 \sin^2 \theta \cos \theta$$

$$\frac{dy}{dx} = \frac{3 \cos \theta - 6 \sin^2 \theta \cos \theta}{-3 \sin \theta + 6 \cos^2 \theta \sin \theta}$$

$$= \frac{\cos \theta - 2 \sin^2 \theta \cos \theta}{-\sin \theta + 2 \cos^2 \theta \sin \theta} = \cot \theta$$

23. Let $f(x)$ be a polynomial function of second degree. If $f(1) = f(-1)$ and a, b, c are in A.P. then $f'(a), f'(b)$ and $f'(c)$ are in : **[AIEEE-2003]**

- (A) Arithmetic - Geometric Progression (B) A.P.
(C) G.P. (D) H.P.

माना कि $f(x)$ द्विघात का बहुपदीय फलन है यदि $f(1) = f(-1)$ तथा a, b, c स. श्रे. में हो तो $f'(a), f'(b)$ एवं $f'(c)$ होंगे :

[AIEEE-2003]

- (A) समान्तरीय गुणोत्तर श्रेणी में (B) समान्तर श्रेणी में
(C) गुणोत्तर श्रेणी में (D) हरात्मक श्रेणी में

Ans. B

Sol. Let $f(x) = a_1 x^2 + a_2 x + a_3$

$$\therefore f(1) = f(-1)$$

$$\Rightarrow a_1 + a_2 + a_3 = a_1 - a_2 + a_3 \Rightarrow a_2 = 0 \Rightarrow f(x) = a_1 x^2 + a_3$$

$$f'(x) = 2a_1 x$$

$$f'(a) = 2a_1 a$$

$$f'(b) = 2a_1 b$$

$$f'(c) = 2a_1 c$$

$$\therefore a, b, c \text{ are in AP}$$

$$\Rightarrow f'(a), f'(b), f'(c) \text{ are also in AP}$$

24. If $x = e^{y+e^{y+\dots\text{to } \infty}}$, $x > 0$, then $\frac{dy}{dx}$ is equal to :

[AIEEE-2004]

यदि $x = e^{y+e^{y+\dots\text{to } \infty}}$, $x > 0$, तो $\frac{dy}{dx}$ बराबर होगा :

[AIEEE-2004]

(A) $\frac{x}{1+x}$

(B) $\frac{1}{x}$

(C) $\frac{1-x}{x}$

(D) $\frac{1+x}{x}$

Ans. C

Sol. $x = e^{y+x}$

$$x + y = \ln x$$

$$1 + \frac{dy}{dx} = \frac{1}{x}$$

$$\frac{dy}{dx} = \frac{1-x}{x}$$

25. If $x^m \cdot y^n = (x+y)^{m+n}$, then $\frac{dy}{dx}$ is equal to :

[AIEEE-2006]

यदि $x^m \cdot y^n = (x+y)^{m+n}$, तो $\frac{dy}{dx}$ बराबर होगा :

[AIEEE-2006]

(A) $\frac{x+y}{xy}$

(B) xy

(C) $\frac{x}{y}$

(D) $\frac{y}{x}$

Ans. D

Sol. $x^m y^n = (x+y)^{m+n}$

taking log both sides

$$m \ln x + n \ln y = (m+n) \cdot \ln(x+y)$$

$$\frac{m}{x} + \frac{n}{y} \frac{dy}{dx} = \frac{(m+n)}{x+y} \left(1 + \frac{dy}{dx} \right)$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{n}{y} - \frac{(m+n)}{x+y} \right) = \frac{m+n}{x+y} - \frac{m}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x}$$

26. $y = \frac{1}{1 + (\tan \theta)^{\sin \theta - \cos \theta} + (\cot \theta)^{\cos \theta - \cot \theta}} + \frac{1}{1 + (\tan \theta)^{\cos \theta - \sin \theta} + (\cot \theta)^{\sin \theta - \cot \theta}} + \frac{1}{1 + (\tan \theta)^{\cos \theta - \cot \theta} + (\cot \theta)^{\cot \theta - \sin \theta}}$

then $\frac{dy}{d\theta}$ at $\theta = \pi/3$ is :

- (A) 0 (B) 1 (C) $\sqrt{3}$ (D) None of these

$$y = \frac{1}{1 + (\tan \theta)^{\sin \theta - \cos \theta} + (\cot \theta)^{\cos \theta - \cot \theta}} + \frac{1}{1 + (\tan \theta)^{\cos \theta - \sin \theta} + (\cot \theta)^{\sin \theta - \cot \theta}} + \frac{1}{1 + (\tan \theta)^{\cos \theta - \cot \theta} + (\cot \theta)^{\cot \theta - \sin \theta}}$$

है तो $\frac{dy}{d\theta}$ का मान $\theta = \pi/3$ पर होगा :

- (A) 0 (B) 1 (C) $\sqrt{3}$ (D) इनमें से कोई नहीं

Ans. A

Sol. $y = \frac{1}{1 + (\tan \theta)^{\sin \theta - \cos \theta} + (\cot \theta)^{\cos \theta - \cot \theta}} + \frac{1}{1 + (\tan \theta)^{\cos \theta - \sin \theta} + (\cot \theta)^{\sin \theta - \cot \theta}} + \frac{1}{1 + (\tan \theta)^{\cos \theta - \cot \theta} + (\cot \theta)^{\cot \theta - \sin \theta}}$

$$= \frac{1}{1 + \frac{(\tan \theta)^{\sin \theta}}{(\tan \theta)^{\cos \theta}} + \frac{(\cot \theta)^{\cos \theta}}{(\cot \theta)^{\cot \theta}}} + \frac{1}{1 + \frac{(\tan \theta)^{\cos \theta}}{(\tan \theta)^{\sin \theta}} + \frac{(\cot \theta)^{\sin \theta}}{(\cot \theta)^{\cot \theta}}} + \frac{1}{1 + \frac{(\tan \theta)^{\cos \theta}}{(\tan \theta)^{\cot \theta}} + \frac{(\cot \theta)^{\cot \theta}}{(\cot \theta)^{\sin \theta}}}$$

$$\Rightarrow \frac{1}{1 + \frac{(\tan \theta)^{\sin \theta}}{(\tan \theta)^{\cos \theta}} + \frac{(\tan \theta)^{\cot \theta}}{(\tan \theta)^{\cos \theta}}} + \frac{1}{1 + \frac{(\tan \theta)^{\cos \theta}}{(\tan \theta)^{\sin \theta}} + \frac{(\tan \theta)^{\cot \theta}}{(\tan \theta)^{\sin \theta}}} + \frac{1}{1 + \frac{(\tan \theta)^{\cos \theta}}{(\tan \theta)^{\cot \theta}} + \frac{(\tan \theta)^{\sin \theta}}{(\tan \theta)^{\cot \theta}}}$$

$$\Rightarrow \frac{(\tan \theta)^{\cos \theta}}{(\tan \theta)^{\sin \theta} + (\tan \theta)^{\cos \theta} + (\tan \theta)^{\cot \theta}} + \frac{(\tan \theta)^{\sin \theta}}{(\tan \theta)^{\sin \theta} + (\tan \theta)^{\cos \theta} + (\tan \theta)^{\cot \theta}} +$$

$$\frac{(\tan \theta)^{\cot \theta}}{(\tan \theta)^{\sin \theta} + (\tan \theta)^{\cos \theta} + (\tan \theta)^{\cot \theta}}$$

$$y = \frac{(\tan \theta)^{\sin \theta} + (\tan \theta)^{\cos \theta} + (\tan \theta)^{\cot \theta}}{(\tan \theta)^{\sin \theta} + (\tan \theta)^{\cos \theta} + (\tan \theta)^{\cot \theta}}$$

$$y = 1$$

$$\frac{dy}{dx} = 0$$

27. Let $f'(x) = \sin(x^2)$ and $y = f(x^2 + 1)$ then $\frac{dy}{dx}$ at $x = 1$ is :

माना $f'(x) = \sin(x^2)$ तथा $y = f(x^2 + 1)$ है, तो $\frac{dy}{dx}$ का मान $x = 1$ पर होगा :

- (A) $2 \sin 2$ (B) $2 \cos 2$ (C) $2 \sin 4$ (D) $\cos 2$

Ans. C

Sol. $f'(x) = \sin(x^2)$ $y = f(x^2 + 1)$

$$y = f(x^2 + 1)$$

$$\frac{dy}{dx} = f'(x^2 + 1) \cdot 2x$$

$$\left. \frac{dy}{dx} \right|_{x=1} = f'(1+1) \cdot 2(1)$$

$$= 2f'(2)$$

$$f'(2) = \sin(2^2) = \sin 4$$

$$\left. \frac{dy}{dx} \right|_{x=1} = 2 \sin 4$$

28. If $f(x) = |\sin x - |\cos x||$, then $f'\left(\frac{7\pi}{6}\right) =$

यदि $f(x) = |\sin x - |\cos x||$ है, तो $f'\left(\frac{7\pi}{6}\right) =$

- (A) $\frac{\sqrt{3}+1}{2}$ (B) $\frac{1-\sqrt{3}}{2}$ (C) $\frac{\sqrt{3}-1}{2}$ (D) $\frac{-1-\sqrt{3}}{2}$

Ans. C

Sol. $f(x) = |\sin x - |\cos x||$

$$x = \frac{7\pi}{6}$$

$$= |\sin x + \cos x|$$

$$f(x) = -\sin x - \cos x$$

$$f'(x) = -\cos x + \sin x$$

$$x = \frac{7\pi}{6}$$

$$f'(x) = -\cos \frac{7\pi}{6} + \sin \frac{7\pi}{6}$$

$$f'\left(\frac{7\pi}{6}\right) = \frac{\sqrt{3}-1}{2}$$

29.. If $f(x) = \sqrt{\frac{1+\sin^{-1}x}{1-\tan^{-1}x}}$; then $f'(0)$ is equal to :

यदि $f(x) = \sqrt{\frac{1+\sin^{-1}x}{1-\tan^{-1}x}}$ है तो $f'(0)$ बराबर होगा :

(A) 4

(B) 3

(C) 2

(D) 1

Ans. D

Sol. $f(x) = \sqrt{\frac{1+\sin^{-1}x}{1-\tan^{-1}x}}$

$$f'(x) = \frac{1}{2\sqrt{\frac{1+\sin^{-1}x}{1-\tan^{-1}x}}} \times \frac{\frac{1}{\sqrt{1-x^2}} \times (1-\tan^{-1}x) + \frac{1}{1+x^2} (1+\sin^{-1}x)}{(1-\tan^{-1}x)^2}$$

$$x = 0$$

$$f'(0) = 1$$

30. If $f(x) = 2 + |x| - |x-1| - |x+1|$, then $f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) + f'\left(\frac{5}{2}\right)$ is equal to :

यदि $f(x) = 2 + |x| - |x-1| - |x+1|$ है, तो $f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) + f'\left(\frac{5}{2}\right)$ का मान होगा :

(A) 1

(B) -1

(C) 2

(D) -2

Ans. D

Sol. $f(x) = 2 + |x| - |x-1| - |x+1|$

$$x = -\frac{1}{2}$$

$$f(x) = 2 - x - x + x - x - 1$$

$$f(x) = 1 - 2x$$

$$f'(x) = -2 = f'\left(-\frac{1}{2}\right) = -2$$

$$x = \frac{1}{2}$$

$$f(x) = 2 + \frac{1}{2} - x + 1 - x - 1$$

$$f(x) = \frac{5}{2} - 2x = f'(x) = -2$$

$$f'\left(\frac{1}{2}\right) = 2$$

$$x = \frac{3}{2}, \frac{5}{2}$$

$$f(x) = 2 + x - x + 1 - x - 1$$

$$f(x) = 2 - x$$

$$f'(x) = -1$$

$$f'\left(\frac{3}{2}\right) = f'\left(\frac{5}{2}\right) = -1$$

$$f'\left(-\frac{1}{2}\right) + f'\left(\frac{1}{2}\right) + f'\left(\frac{3}{2}\right) + f'\left(\frac{5}{2}\right)$$

$$= -2$$

31. Let $g(x) = \ln f(x)$ where $f(x)$ is a twice differentiable positive function on $(0, \infty)$ such that

$$f(x+1) = x f(x). \text{ Then for } N = 1, 2, 3, \dots \quad g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) = \quad \text{[JEE-2008]}$$

माना $g(x) = \ln f(x)$ है जहाँ $f(x)$ अन्तराल $(0, \infty)$ में दो बार अवकलनीय धनात्मक फलन इस प्रकार है कि

$$f(x+1) = x f(x) \text{ तो } N = 1, 2, 3, \dots \text{ के लिये } g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) = \quad [\text{JEE-2008}]$$

- (A) $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$ (B) $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right\}$
 (C) $-4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$ (D) $4 \left\{ 1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N+1)^2} \right\}$

Ans. A

Sol. $g(x+1) = \log [f(x+1)] = \log [xf(x)] = \log x + \log [f(x)] = \log x + g(x)$

$$\Rightarrow g(x+1) - g(x) = \log x$$

$$\Rightarrow g''(x+1) - g''(x) = -\frac{1}{x^2}$$

$$\Rightarrow g''\left(1 + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) = -4$$

$$g''\left(1 + \frac{3}{2}\right) - g''\left(\frac{3}{2}\right) = -\frac{4}{9}$$

.....

$$g''\left(N + \frac{1}{2}\right) - g''\left(N - \frac{1}{2}\right) = -\frac{4}{(2N-1)^2}$$

$$\text{Adding all, } g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) = -4 \left\{ 1 + \frac{1}{9} + \dots + \frac{1}{(2N-1)^2} \right\}$$

32. Let $f: (-1, 1) \rightarrow \mathbb{R}$ be a differentiable function with $f(0) = -1$, $f'(0) = 1$. Let $g(x) = |f(2f(x) + 2)|^2$.] Then $g'(0)$:
[AIEEE-2010]

माना $f: (-1, 1) \rightarrow \mathbb{R}$ एक ऐसा अवकलनीय फलन है जिसमें $f(0) = -1$, $f'(0) = 1$ है। माना $g(x) = |f(2f(x) + 2)|^2$ । तो $g'(0)$ बराबर
[AIEEE-2010]

- (A) 4 (B) -4 (C) 0 (D) -2

Ans. B

Sol. $g'(x) = 2f(2f(x) + 2) \cdot f'(2f(x) + 2) \cdot 2f'(x)$

$$g'(0) = 2f(2f(0) + 2) \cdot f'(2f(0) + 2) \cdot 2f'(0)$$

$$= 2f(0) f'(0) 2f'(0) = (2) (-1) (1) (2) (1) = -4$$

Hence correct option is (1)

अतः सही विकल्प (1) है।

33. For $x \in \mathbb{R}$, $f(x) = |\log 2 - \sin x|$ and $g(x) = f(f(x))$, then :

[JEE(Main)-2016]

- (A) $g'(0) = -\cos(\log 2)$
- (B) g is differentiable at $x = 0$ and $g'(0) = -\sin(\log 2)$
- (C) g is not differentiable at $x = 0$
- (D) $g'(0) = \cos(\log 2)$

$x \in \mathbb{R}$ के लिए, $f(x) = |\log 2 - \sin x|$ तथा $g(x) = f(f(x))$ हैं, तो :

[JEE(Main)-2016]

- (A) $g'(0) = -\cos(\log 2)$
- (B) $x = 0$ पर g अवकलनीय है तथा $g'(0) = -\sin(\log 2)$ है
- (C) $x = 0$ पर g अवकलनीय नहीं है
- (D) $g'(0) = \cos(\log 2)$

Ans. D

Sol. $g(x) = |\log 2 - \sin |\log 2 - \sin x||$
 $g(x)$ near by of $x = 0$
 $g(x) = \log 2 - \sin(\log 2 - \sin x)$
 $g'(x) = -(\cos(\log 2 - \sin x))(-\cos x)$
 $g'(0) = \cos(\log 2)$

34. If for $x \in \left(0, \frac{1}{4}\right)$, the derivative of $\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ is $\sqrt{x} \cdot g(x)$, then $g(x)$ equals :

[JEE(Main)-2017]

यदि $x \in \left(0, \frac{1}{4}\right)$ के लिए $\tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right)$ का अवकलन $\sqrt{x} \cdot g(x)$ है, तो $g(x)$ बराबर है :

[JEE(Main)-2017]

- (A) $\frac{3x}{1-9x^3}$
- (B) $\frac{3}{1+9x^3}$
- (C) $\frac{9}{1+9x^3}$
- (D) $\frac{3x\sqrt{x}}{1-9x^3}$

Ans. C

Sol. $f(x) = \tan^{-1}\left(\frac{6x\sqrt{x}}{1-9x^3}\right) = 2 \tan^{-1}(3x\sqrt{x})$

$$\Rightarrow f'(x) = \frac{9\sqrt{x}}{1+9x^3} = \sqrt{x} \cdot g(x)$$

$$\Rightarrow g(x) = \frac{9}{1+9x^3}$$

DERIVATIVE OF INVERSE OF A FUNCTION

35. Let $g(x)$ be the inverse of $f(x)$ such that $f'(x) = \frac{1}{1+x^5}$, then $\frac{d^2(g(x))}{dx^2}$ is equal to :

माना $f(x)$ का प्रतिलोम फलन $g(x)$ इस प्रकार है कि $f'(x) = \frac{1}{1+x^5}$ है तो $\frac{d^2(g(x))}{dx^2}$ का मान होगा :

- (A) $\frac{1}{1+(g(x))^5}$ (B) $\frac{g'(x)}{1+(g(x))^5}$ (C) $5(g(x))^4(1+(g(x))^5)$ (D) $1+(g(x))^5$

Ans. C

Sol. $f'(x) = \frac{1}{1+x^5}, \quad \frac{d^2(g(x))}{dx^2} = ?$

$$f^{-1}(x) = g(x)$$

$$f(g(x)) = x$$

$$f'(g(x)) \times g'(x) = 1$$

$$g'(x) = \frac{1}{f'(g(x))}$$

$$= \frac{1}{\frac{1}{1+[g(x)]^5}} = 1 + (g(x))^5$$

$$g''(x) = 0 + 5[g(x)]^4 \times g'(x)$$

$$= 5[g(x)]^4 [1 + (g(x))^5]$$

36. Let $f(x) = x + \sin x$. Suppose g denotes the inverse function of f . The value of $g'\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right)$ has the value equal to:

माना $f(x) = x + \sin x$. यदि g फलन f का प्रतिलोम है, तो $g'\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right)$ का मान है :

- (A) $\sqrt{2} - 1$ (B) $\frac{\sqrt{2} + 1}{\sqrt{2}}$ (C) $2 - \sqrt{2}$ (D) $\sqrt{2} + 1$

Ans. C

Sol. $f(x) = y = x + \sin x$

$$\frac{dy}{dx} = 1 + \cos x$$

$$g'(y) = \frac{dx}{dy} = \frac{1}{1 + \cos x}$$

$$\text{where } y = \frac{\pi}{4} + \frac{1}{\sqrt{2}} = x + \sin x \Rightarrow x = \frac{\pi}{4}$$

$$\therefore g'\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right) = \frac{1}{1 + (1/\sqrt{2})} = \frac{\sqrt{2}}{\sqrt{2} + 1} = \sqrt{2}(\sqrt{2} - 1) = 2 - \sqrt{2}]$$

37. Let g is the inverse function of f and $f'(x) = \frac{x^{10}}{(1+x^2)}$. If $g(2) = a$ then $g'(2)$ is equal to :

यदि $g(x)$, f का प्रतिलोम फलन है और $f'(x) = \frac{x^{10}}{(1+x^2)}$. यदि $g(2) = a$ तब $g'(2)$ बराबर :

(A) $\frac{5}{2^{10}}$

(B) $\frac{1+a^2}{a^{10}}$

(C) $\frac{a^{10}}{1+a^2}$

(D) $\frac{1+a^{10}}{a^2}$

Ans. B

Sol. $f[g(x)] = x \Rightarrow f'[g(x)] \cdot [g'(x)] = 1 \Rightarrow f'[g(2)] g'(2) = 1 \Rightarrow f'(a) g'(2) [putting x = 2]$

$$\text{given, } f'(a) = \frac{a^{10}}{1+a^2} \Rightarrow g'(2) = \frac{1+a^2}{a^{10}}]$$

Alternative : $g[f(x)] = x$

$$g'[f(x)] \cdot f'(x) = 1$$

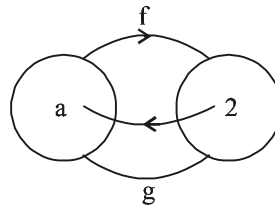
now $g(2) = a \Rightarrow f(a) = 2$

$\therefore$ g and f are inverse of each other

now $f(x) = 2 \Rightarrow g(2) = x = a$

$$\therefore g'(2) \cdot f'(a) = 1$$

$$g'(2) = \frac{1}{f'(a)} = \frac{1+a^2}{a^{10}}$$



HIGHER ORDER DERIVATIVE

38. If $y = \frac{1}{x^2 - a^2}$, then $\frac{d^2y}{dx^2}$ equals :

यदि $y = \frac{1}{x^2 - a^2}$, तो $\frac{d^2y}{dx^2}$ बराबर है :

- (A) $\frac{3x^2 + a^2}{(x^2 - a^2)^3}$ (B) $\frac{3x^2 + a^2}{(x^2 - a^2)^4}$ (C) $\frac{2(3x^2 + a^2)}{(x^2 - a^2)^3}$ (D) $\frac{2(3x^2 + a^2)}{(x^2 - a^2)^4}$

Ans. C

Sol. $y = \frac{1}{x^2 - a^2}$

$$\frac{dy}{dx} = \frac{-1(2x)}{(x^2 - a^2)^2}$$

$$\frac{d^2y}{dx^2} = \frac{-2(x^2 - a^2)^2 + 2x \times 2(x^2 - a^2)}{(x^2 - a^2)^4} \times 2x$$

$$= \frac{2(x^2 - a^2)[4x^2 - x^2 + a^2]}{(x^2 - a^2)^4}$$

$$= \frac{2(3x^2 + a^2)}{(x^2 - a^2)^3}$$

39. If $y = ae^{\alpha x} - be^{-\alpha x}$, then the value of $\frac{d^2y}{dx^2}$ equals :

यदि $y = ae^{\alpha x} - be^{-\alpha x}$, तब $\frac{d^2y}{dx^2}$ का मान बराबर है :

- (A) αy (B) $-\alpha y$ (C) $\alpha^2 y$ (D) $-\alpha^2 y$

Ans. C

Sol. $y = ae^{\alpha x} - be^{-\alpha x}$

$$\frac{dy}{dx} = a\alpha e^{\alpha x} + b\alpha e^{-\alpha x}$$

$$\frac{d^2y}{dx^2} = a\alpha^2 e^{\alpha x} - b\alpha^2 e^{-\alpha x}$$

$$\frac{d^2y}{dx^2} = \alpha^2 y$$

40. If $y^{1/m} = x + \sqrt{1+x^2}$, then $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ equals :

- (A) $\frac{1}{m} y^2$ (B) $m^2 y$ (C) $\frac{1}{m^2} y^2$ (D) None of these

यदि $y^{1/m} = x + \sqrt{1+x^2}$, तब $(1+x^2)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ बराबर है :

- (A) $\frac{1}{m} y^2$ (B) $m^2 y$ (C) $\frac{1}{m^2} y^2$ (D) इनमें से कोई नहीं

Ans. B

Sol. $y^{\frac{1}{m}} = x + \sqrt{1+x^2}$

$$\frac{1}{m} \ln y = \ln(x + \sqrt{1+x^2})$$

$$\frac{1}{my} \frac{dy}{dx} = \frac{1 + \frac{2x}{2\sqrt{1+x^2}}}{x + \sqrt{1+x^2}}$$

$$\frac{dy}{dx} = \frac{my}{\sqrt{1+x^2}}$$

$$\frac{d^2y}{dx^2} = \frac{m \frac{dy}{dx} \sqrt{1+x^2} - my \frac{2x}{2\sqrt{1+x^2}}}{1+x^2}$$

$$(1+x^2)\frac{d^2y}{dx^2} = m\sqrt{1+x^2} \frac{dy}{dx} - \frac{mxy}{\sqrt{1+x^2}}$$

$$(1+x^2)\frac{d^2y}{dx^2} = m\sqrt{1+x^2} \frac{dy}{dx} - x \frac{dy}{dx}$$

$$(1+x^2) + x \frac{dy}{dx} = m\sqrt{1+x^2} \left(\frac{my}{\sqrt{1+x^2}} \right)$$

$$= m^2 y$$

41. If $y = a \cos(\log x) + b \sin(\log x)$ then $(x^2 y_2 + xy_1)$ equals to :

यदि $y = a \cos(\log x) + b \sin(\log x)$ तब $(x^2 y_2 + xy_1)$ बराबर है :

(A) y (B) ay (C) by (D) -y

Ans. D

Sol. $y = a \cos(\log x) + b \sin(\log x)$

$$y_1 = \frac{-a \sin(\log x)}{x} + \frac{b \cos(\log x)}{x}$$

$$y_1 = \frac{-a \sin(\log x) + b \cos(\log x)}{x}$$

$$y_2 = \frac{x \left(\frac{-a \cos(\log x)}{x} - \frac{b \sin(\log x)}{x} \right) - (b \cos(\log x) - a \sin(\log x))}{x^2}$$

$$x^2 y_2 = a \cos(\log x) - b \sin(\log x) - b \cos(\log x) + a \sin(\log x)$$

$$xy_1 = -a \sin(\log x) + b \cos(\log x)$$

$$x^2 y_2 + xy_1 = -a \cos(\log x) - b \sin(\log x)$$

$$= -y$$

42. If $y = \sin 2x \cos 6x$, then the value of $\frac{d^2 y}{dx^2}$ is :

(A) $8 \sin 4x + 32 \sin 8x$ (B) $32 \sin 8x - 8 \sin 4x$ (C) $8 \sin 4x - 32 \sin 8x$ (D) None of these

यदि $y = \sin 2x \cos 6x$, तब $\frac{d^2 y}{dx^2}$ का मान होगा :

(A) $8 \sin 4x + 32 \sin 8x$ (B) $32 \sin 8x - 8 \sin 4x$ (C) $8 \sin 4x - 32 \sin 8x$ (D) इनमें से कोई नहीं

Ans. C

Sol. $y = \frac{2 \sin 2x \cos 6x}{2}$

$$y = \frac{\sin 8x - \sin 4x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2}[8 \cos 8x - 4 \cos 4x]$$

$$= 4 \cos 8x - 2 \cos 4x$$

$$\frac{d^2y}{dx^2} = -32 \sin 8x + 8 \sin 4x$$

43. If $y = \sqrt{2x - x^2}$, then $yy_2 + y_1^2$ is equal to :

- (A) 0 (B) -1 (C) 1 (D) Does not exist

यदि $y = \sqrt{2x - x^2}$ है तो $yy_2 + y_1^2$ बराबर होगा :

- (A) 0 (B) -1 (C) 1 (D) विद्यमान नहीं

Ans. B

Sol. $y = \sqrt{2x - x^2}$

$$y_1 = \frac{2 - 2x}{2\sqrt{2x - x^2}} = \frac{1 - x}{\sqrt{2x - x^2}}$$

$$y_2 = \frac{(-1)\sqrt{2x - x^2} - (1 - x) \frac{2 - 2x}{2\sqrt{2x - x^2}}}{(2x - x^2)}$$

$$y_2 = \frac{-\sqrt{2x - x^2} - \frac{(1 - x)^2}{\sqrt{2x - x^2}}}{2x - x^2}$$

$$y_2 = \frac{-2x + x^2 - 1 - x^2 + 2x}{(2x - x^2)\sqrt{2x - x^2}}$$

$$y_2 = \frac{1}{(x^2 - 2x)\sqrt{2x - x^2}}$$

$$yy_2 + y_1^2 = \frac{\sqrt{2x - x^2}}{(x^2 - 2x)\sqrt{2x - x^2}} + \frac{(1 - x)^2}{2x - x^2}$$

$$= \frac{1 - (1-x)^2}{x^2 - 2x} = -1$$

44. If $f(x) = x^n$, then the value of $f(1) - \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + \dots + \frac{(-1)^n f^{(n)}(1)}{n!}$ is :

यदि $f(x) = x^n$ तो $f(1) - \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \frac{f'''(1)}{3!} + \dots + \frac{(-1)^n f^{(n)}(1)}{n!}$ का मान है :

- (A) 1 (B) 2^n (C) 2^{n-1} (D) 0

Ans. D

Sol. $f(x) = x^n$

$$\therefore f'(x) = nx^{n-1}, f''(x) = n(n-1)x^{n-2}$$

.....

$$f^{(n)}(x) = n!$$

$$\text{Now } f(1) - \frac{f'(1)}{1!} + \frac{f''(1)}{2!} - \dots - \frac{f^{(n)}(1)}{n!}$$

$$= 1 - \frac{n}{1!} + \frac{n(n-1)}{2!} - \dots - \frac{n!}{n!} = (1-1)^n = 0$$

45. Let 'f' be a differentiable real valued function satisfying $f(x+2y) = f(x) + f(2y) + 6xy(x+2y) \forall x, y \in \mathbb{R}$. Then $f''(0), f''(1), f''(2) \dots$ are in :

- (A) AP (B) GP (C) HP (D) None of these

माना 'f' एक अवकलनीय वास्तविक मान फलन इस प्रकार है कि $f(x+2y) = f(x) + f(2y) + 6xy(x+2y) \forall x, y \in \mathbb{R}$ है तो $f''(0), f''(1), f''(2) \dots$ में होंगे :

- (A) AP (B) GP (C) HP (D) इनमें से कोई नहीं

Ans. A

Sol. $x = y = 0$

$$f(0) = 0$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{[f(x) + f(h) + 3xh(x+h) - f(x)]}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{f(h)}{h} + 3x(x+h) \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{f'(h)}{1} \right] + 3x^2$$

$$f'(x) = f'(0) + 3x^2$$

$$f''(x) = 0 + 6x$$

$$f''(0), f''(1), f''(2)$$

$$0, 6, 12 \rightarrow \text{A.P.}$$

46. $\frac{d^2x}{dy^2}$ equals : [JEE-2007]

$\frac{d^2x}{dy^2}$ बराबर होगा : [JEE-2007]

(A) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (B) $-\left(\frac{d^2y}{dx^2}\right)^{-1} \left(\frac{dy}{dx}\right)^{-3}$ (C) $\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-2}$ (D) $-\left(\frac{d^2y}{dx^2}\right) \left(\frac{dy}{dx}\right)^{-3}$

Ans. D

Sol. $\frac{d}{dy} \left(\frac{dx}{dy} \right) = \frac{d}{dy} \left(\frac{1}{dy/dx} \right)$

$$= - \left(\frac{dy}{dx} \right)^{-2} \frac{d^2y}{dx^2} \cdot \frac{dx}{dy} = - \frac{d^2y}{dx^2} \left(\frac{dy}{dx} \right)^{-3}$$

L'HOSPITAL RULE

47. Let $f(x) = \frac{e^{\tan x} - e^x + \ln(\sec x + \tan x) - x}{\tan x - x}$ be a continuous function at $x = 0$. The value of $f(0)$ equals :

माना $f(x) = \frac{e^{\tan x} - e^x + \ln(\sec x + \tan x) - x}{\tan x - x}$, $x = 0$ पर सतत् फलन है तब $f(0)$ बराबर :

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 2

Ans. C

Sol. For continuity of f at $x = 0$, we have

$$\begin{aligned} k = f(0) &= \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{e^{\tan x} - e^x}{\tan x - x} + \lim_{x \rightarrow 0} \frac{\ln(\sec x + \tan x) - x}{\left(\frac{\tan x - x}{x^3}\right)x^3} \\ &= \lim_{x \rightarrow 0} \frac{e^x(e^{\tan x - x} - 1)}{\tan x - x} + 3 \lim_{x \rightarrow 0} \frac{\ln(\sec x + \tan x) - x}{x^3} = 1 + 3 \lim_{x \rightarrow 0} \frac{\sec x - 1}{3x^2} \quad (\text{Using L.H. Rule}) \\ &= 1 + \frac{1}{2} = \frac{3}{2} \quad \text{Ans.} \end{aligned}$$

48. If the normal of $y = f(x)$ at $(0, 0)$ is given by $y - x = 0$, then $\lim_{x \rightarrow 0} \frac{x^2}{f(x^2) - 20f(9x^2) + 2f(99x^2)}$ is equal to :

- (A) $\frac{1}{19}$ (B) $-\frac{1}{19}$ (C) $\frac{1}{2}$ (D) non-existent

यदि $y = f(x)$ का $(0, 0)$ अभिलम्ब समीकरण $y - x = 0$ है, तो $\lim_{x \rightarrow 0} \frac{x^2}{f(x^2) - 20f(9x^2) + 2f(99x^2)}$ बराबर होगी :

- (A) $\frac{1}{19}$ (B) $-\frac{1}{19}$ (C) $\frac{1}{2}$ (D) विद्यमान नहीं

Ans. B

Sol. The form being $\frac{0}{0}$, so the required limit by L'Hospital rule is

$$\lim_{x \rightarrow 0} \frac{2x}{2x f'(x^2) - 360x f'(9x^2) + 396x f'(99x^2)} = \lim_{x \rightarrow 0} \frac{1}{f'(x^2) - 180f'(9x^2) + 198f'(99x^2)}$$

For $f(x) = 0$, the slope of tangent is $f'(x)$ and slope of normal at $x = 0$ is $\frac{-1}{f'(0)} = 1$ (Given)

$$\Rightarrow f'(0) = -1$$

$$\therefore \text{Limit} = \frac{1}{f'(0) - 180f'(0) + 198f'(0)} = \frac{-1}{(1 - 180 + 198)} = \frac{-1}{19} \quad \text{Ans.}$$

49. $\lim_{x \rightarrow \infty} \left(\frac{x^{100}}{e^x} + \left(\cos \frac{2}{x} \right)^{x^2} \right)$ equals :

$$\lim_{x \rightarrow \infty} \left(\frac{x^{100}}{e^x} + \left(\cos \frac{2}{x} \right)^{x^2} \right) \text{ बराबर होगा :}$$

- (A) e^{-1} (B) e^{-4} (C) $(1 + e^{-2})$ (D) e^{-2}

Ans. D

Sol. $\lim_{x \rightarrow \infty} \frac{x^{100}}{e^x} = 0$ (using L'Hospital's Rule)

$$\lim_{x \rightarrow \infty} \left(\cos \frac{2}{x} \right)^{x^2} \quad (1^\infty)$$

$$= e^{\lim_{x \rightarrow \infty} x^2 \left(\cos \frac{2}{x} - 1 \right)} = e^{-\lim_{x \rightarrow 0} \left(\frac{1 - \cos t}{t^2} \right) \cdot 4} ; \text{ put } \frac{2}{x} = t \Rightarrow x = \frac{2}{t}$$

$$= e^{-2} \text{ Ans.}$$

50. Let $f(x)$ be differentiable at $x = h$ then $\lim_{x \rightarrow h} \frac{(x+h)f(x) - 2hf(h)}{x-h}$ is equal to :

माना $f(x)$; $x = h$ पर अवकलनीय है, तब $\lim_{x \rightarrow h} \frac{(x+h)f(x) - 2hf(h)}{x-h}$ बराबर :

- (A) $f(h) + 2hf'(h)$ (B) $2f(h) + hf'(h)$ (C) $hf(h) + 2f'(h)$ (D) $hf(h) - 2f'(h)$

Ans. A

Sol. Use L'Hospital's rule

51. For $x > 0$, $\lim_{x \rightarrow 0} \left((\sin x)^{1/x} + (1/x)^{\sin x} \right)$ is :

$x > 0$ के लिये $\lim_{x \rightarrow 0} \left((\sin x)^{1/x} + (1/x)^{\sin x} \right)$ है :

- (A) 0 (B) -1 (C) 1 (D) 2

Ans. C

Sol. for $x > 0$

$$\lim_{x \rightarrow 0} \left((\sin x)^{\frac{1}{x}} + \left(\frac{1}{x} \right)^{\sin x} \right)$$

$$\begin{aligned}
& \lim_{x \rightarrow 0} (\sin x)^{\frac{1}{x}} + \lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{\sin x} = 0 + \lim_{x \rightarrow 0} \left(\frac{1}{x} \right)^{\sin x} = e^{\lim_{x \rightarrow 0} (\sin x) \cdot \ln \left(\frac{1}{x} \right)} = e^{\lim_{x \rightarrow 0} \frac{\ln(x)}{-\operatorname{cosec} x}} \left(\frac{\infty}{\infty} \text{ form} \right) \\
& = e^{\lim_{x \rightarrow 0} \frac{\frac{1}{x}}{(\operatorname{cosec} x \cot x)}} = e^{\lim_{x \rightarrow 0} \frac{\sin^2 x}{x \cos x}} = e^0 = 1
\end{aligned}$$

Exercise J-II

BASED ON DIFFERENTIATION TECHNIQUES

1. Let f, g and h are differentiable functions. If $f(0) = 1; g(0) = 2; h(0) = 3$ and the derivatives of their pair wise products at $x = 0$ are $(fg)'(0) = 6; (gh)'(0) = 4$ and $(hf)'(0) = 5$ then compute the value of $(fgh)'(0)$.

माना f, g तथा h अवकलनीय फलन है। यदि $f(0) = 1; g(0) = 2; h(0) = 3$ तथा इनके युग्मों के गुणनों के $x = 0$ पर अवकलज इस प्रकार है। $(fg)'(0) = 6; (gh)'(0) = 4$ और $(hf)'(0) = 5$ तो $(fgh)'(0)$ का मान निकालिए।

Ans. 16

Sol. $(fgh)' = \frac{(fg)'h + f(gh)' + g(fh)'}{2}$

$$(fgh)'(0) = \frac{6 \times 3 + 1 \times 4 + 2 \times 5}{2}$$

$$= 16$$

2. (a) If $y = (\cos x)^{\ln x} + (\ln x)^x$ find $\frac{dy}{dx}$

- (b) If $y = e^{x^{e^x}} + e^{x^{x^e}} + x^{e^{e^x}}$. Find $\frac{dy}{dx}$.

∴ यदि $y = (\cos x)^{\ln x} + (\ln x)^x$ तो $\frac{dy}{dx}$ ज्ञात कीजिए

(b) यदि $y = e^{x^{e^x}} + e^{x^{x^e}} + x^{e^{e^x}}$ तो $\frac{dy}{dx}$ ज्ञात कीजिए।

Ans. (a) $\frac{dy}{dx} = (\cos x)^{\ln x} \left[\frac{\ln(\cos x)}{x} - \tan x \ln x \right] + (\ln x)^x \left[\frac{1}{\ln x} + \ln(\ln x) \right]$

(b) $\frac{dy}{dx} = e^{x^{e^x}} \cdot x^{e^x} \left[\frac{e^x}{x} + e^x \ln x \right] + e^{x^{x^e}} x^{e-1} x^{x^e} [1 + e \ln x] + x^{e^{e^x}} e^{e^x} \left[\frac{1}{x} + e^x \ln x \right]$

Sol. (a) $y = (\cos x)^{\ln x} + (\ln x)^x$

$$y_1 = (\cos x)^{\ln x}$$

$$\log y_1 = \ln x \ln(\cos x)$$

$$\frac{1}{y_1} \frac{dy_1}{dx} = \frac{\ln(\cos x)}{x} - \ln x \frac{\sin x}{\cos x}$$

$$\frac{dy_1}{dx} = (\cos x)^{\ln x} \left[\frac{\ln(\cos x)}{x} - \tan x \ln x \right]$$

$$y_2 = (\ln x)^x$$

$$\log y_2 = x \ln(\ln x)$$

$$\frac{1}{y_2} \frac{dy_2}{dx} = \ln(\ln x) + \frac{1}{\ln x}$$

$$\frac{dy_2}{dx} = (\ln x)^x \left[\ln(\ln x) + \frac{1}{\ln x} \right]$$

$$\frac{dy}{dx} = \frac{dy_1}{dx} + \frac{dy_2}{dx}$$

$$(b) \quad y = e^{x^{e^x}} + e^{x^{x^e}} + x^{e^{e^x}}$$

$$y_1 = e^{x^{e^x}}$$

$$\log y_1 = x^{e^x}$$

$$\log(\log y_1) = e^x \ln x$$

$$\frac{1}{y_1} \frac{1}{\log y_1} \frac{dy_1}{dx} = e^x \ln x + \frac{e^x}{x}$$

$$\frac{dy_1}{dx} = e^{x^{e^x}} x^{e^x} \left[e^x \ln x + \frac{e^x}{x} \right]$$

$$y_2 = e^{x^{x^e}}$$

$$\log y_2 = x^{x^e}$$

$$\log(\log y_2) = x^e \ln x$$

$$\frac{1}{y_2} \frac{dy_2}{dx} = ex^{e-1} \ln x + \frac{x^e}{x}$$

$$\frac{dy_2}{dx} = e^{x^{x^e}} \cdot x^{x^e} \left[ex^{e-1} \ln x + \frac{x^e}{x} \right]$$

$$\frac{dy_2}{dx} = e^{x^{x^e}} \cdot x^{x^e} \cdot x^{e-1} [e \ln x + 1]$$

$$y_3 = x^{e^{e^x}}$$

$$\log y_3 = e^{e^x} \log x$$

$$\log (\log y_3) = e^x + \log (\log x)$$

$$\frac{1}{y_3 \log y_3} \frac{dy_3}{dx} = e^x + \frac{1}{x \ln x}$$

$$\frac{dy_3}{dx} = x^{e^{e^x}} \cdot e^{e^x} \log x \left[e^x + \frac{1}{x \ln x} \right]$$

$$\frac{dy_3}{dx} = x^{e^{e^x}} \cdot e^{e^x} \left[\ln x e^x + \frac{1}{x} \right]$$

$$\frac{dy}{dx} = e^{x^{e^x}} \cdot x^{e^x} \left[e^x \ln x + \frac{e^x}{x} \right] + e^{x^{e^x}} \cdot x^{e^x} \cdot x^{e-1} [e \ln x + 1] + x^{e^{e^x}} \cdot e^{e^x} \left[\ln x e^x + \frac{1}{x} \right]$$

3. If $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \ln\sqrt{x+\sqrt{x^2+1}}$ prove that $2y = xy' + \ln y'$. where ' denotes the derivative.

यदि $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \ln\sqrt{x+\sqrt{x^2+1}}$ तो सिद्ध कीजिये $2y = xy' + \ln y'$ जहाँ y' , y का अवकलज है।

Ans.

Sol. $y = \frac{x^2}{2} + \frac{1}{2}x\sqrt{x^2+1} + \ln\sqrt{x+\sqrt{x^2+1}}$

$$2y = x^2 + x\sqrt{x^2+1} + \ln(x + \sqrt{x^2+1}) \quad \dots(i)$$

$$2y' = 2x + \sqrt{x^2+1} + \frac{2x^2}{2\sqrt{x^2+1}} + \frac{1 + \frac{2x}{2\sqrt{x^2+1}}}{x + \sqrt{x^2+1}}$$

$$2y' = 2x + \sqrt{x^2+1} + \frac{x^2}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}}$$

$$= 2x + \sqrt{x^2+1} + \sqrt{x^2+1}$$

$$y' = (x + \sqrt{x^2+1})$$

Put the value of $x + \sqrt{x^2+1}$ in (i)

$$2y = xy' + \ln y'$$

4. If $y = \ln(x^{e^x \cdot a^y})^{y^x}$ find $\frac{dy}{dx}$.

यदि $y = \ln(x^{e^x \cdot a^y})^{y^x}$ तो $\frac{dy}{dx}$ ज्ञात कीजिए।

Ans. $\frac{y}{x} \cdot \frac{x \ln x + x \ln x \cdot \ln y + 1}{\ln x (1 - x - y \ln a)}$

Sol. $y = \ln(x^{e^x \cdot a^y})^{y^x}$

$$y = e^x a^y y^x \ln x$$

$$\ln y = x + y \ln a + x \ln y + \ln(\ln x)$$

$$\frac{1}{y} y' = 1 + y' \ln a + \ln y + \frac{x}{y} y' + \frac{1}{x \ln x}$$

$$\frac{y'}{y} - y'(\ln a) - \frac{x}{y} y' = 1 + \ln y + \frac{1}{x \ln x}$$

$$y' \left(\frac{1 - y \ln a - x}{y} \right) = \frac{x \ln x + x \ln x \ln y + 1}{x \ln x}$$

$$\frac{dy}{dx} = \frac{y}{x} \cdot \frac{x \ln x + x \ln x \ln y + 1}{\ln x (1 - y \ln a - x)}$$

5. If $y = 1 + \frac{x_1}{x - x_1} + \frac{x_2 \cdot x}{(x - x_1)(x - x_2)} + \frac{x_3 \cdot x^2}{(x - x_1)(x - x_2)(x - x_3)} + \dots$ upto $(n + 1)$ terms then prove that

$$\frac{dy}{dx} = \frac{y}{x} \left[\frac{x_1}{x_1 - x} + \frac{x_2}{x_2 - x} + \frac{x_3}{x_3 - x} + \dots + \frac{x_n}{x_n - x} \right]$$

यदि $y = 1 + \frac{x_1}{x - x_1} + \frac{x_2 \cdot x}{(x - x_1)(x - x_2)} + \frac{x_3 \cdot x^2}{(x - x_1)(x - x_2)(x - x_3)} + \dots$ $(n + 1)$ पदों तक तो सिद्ध कीजिये

$$\frac{dy}{dx} = \frac{y}{x} \left[\frac{x_1}{x_1 - x} + \frac{x_2}{x_2 - x} + \frac{x_3}{x_3 - x} + \dots + \frac{x_n}{x_n - x} \right]$$

Ans.

Sol. $\because y = 1 + \frac{x_1}{x-x_1} + \frac{x_2x}{(x-x_1)(x-x_2)} + \frac{x_3x}{(x-x_1)(x-x_2)(x-x_3)} + \dots (n+1) \text{ terms}$

$$= \frac{x}{(x-x_1)} + \frac{x_2x}{(x-x_1)(x-x_2)} + \frac{x_3x}{(x-x_1)(x-x_2)(x-x_3)} + \dots (n+1) \text{ terms}$$

$$= \frac{x^2}{(x-x_1)(x-x_2)} + \frac{x_3x}{(x-x_1)(x-x_2)(x-x_3)} + \dots (n+1) \text{ terms}$$

finally we get

$$y = \frac{x^n}{(x-x_1)(x-x_2)(x-x_3)\dots(x-x_n)}$$

Taking $\log_e$ of both sides, we have

$$\ell ny = n \ell nx - \ell n(x-x_1) - \ell n(x-x_2) - \ell n(x-x_3) - \dots - \ell n(x-x_n)$$

Differentiating both sides w.r.t. x , we get

$$\frac{1}{y} \frac{dy}{dx} = \left[\frac{n}{x} - \frac{1}{x-x_1} - \frac{1}{x-x_2} - \frac{1}{x-x_3} - \dots - \frac{1}{x-x_n} \right]$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} \left[n - \frac{x}{x-x_1} - \frac{x}{x-x_2} - \frac{x}{x-x_3} - \dots - \frac{x}{x-x_n} \right]$$

$$= \frac{y}{x} \left[(1+1+1+1+\dots n \text{ times}) - \frac{x}{x-x_1} - \frac{x}{x-x_2} - \frac{x}{x-x_3} - \dots - \frac{x}{x-x_n} \right]$$

$$= \frac{y}{x} \left[\left(1 - \frac{x}{x-x_1} \right) + \left(1 - \frac{x}{x-x_2} \right) + \left(1 - \frac{x}{x-x_3} \right) + \dots + \left(1 - \frac{x}{x-x_n} \right) \right]$$

$$\therefore \frac{dy}{dx} = \frac{y}{x} \left[\left(\frac{-x_1}{x-x_1} \right) + \left(\frac{-x_2}{x-x_2} \right) + \left(\frac{-x_3}{x-x_3} \right) + \dots + \left(\frac{-x_n}{x-x_n} \right) \right]$$

$$= \frac{y}{x} \left[\frac{x_1}{x_1-x} + \frac{x_2}{x_2-x} + \frac{x_3}{x_3-x} + \dots + \frac{x_n}{x_n-x} \right]$$

6. If $x = \operatorname{cosec} \theta - \sin \theta$; $y = \operatorname{cosec}^n \theta - \sin^n \theta$, then show that $(x^2 + 4) \left(\frac{dy}{dx} \right)^2 - n^2 (y^2 + 4) = 0$.

यदि $x = \operatorname{cosec} \theta - \sin \theta$; $y = \operatorname{cosec}^n \theta - \sin^n \theta$, तो दर्शाइए $(x^2 + 4) \left(\frac{dy}{dx} \right)^2 - n^2 (y^2 + 4) = 0$

Ans.

Sol. $x = \operatorname{cosec} \theta - \sin \theta$

$$\frac{dx}{d\theta} = -\operatorname{cosec} \theta \cot \theta - \cos \theta \times \frac{\sin \theta}{\sin \theta}$$

$$\frac{dx}{d\theta} = -\operatorname{cosec} \theta \cot \theta - \cot \theta \sin \theta$$

$$\frac{dx}{d\theta} = -\cot \theta (\operatorname{cosec} \theta + \sin \theta)$$

$$y = \operatorname{cosec}^n \theta - \sin^n \theta$$

$$\frac{dy}{d\theta} = x \operatorname{cosec}^{n-1} \theta (-\operatorname{cosec} \theta \cot \theta) - n \sin^{n-1} \theta \times \cos \theta$$

$$\frac{dy}{d\theta} = -n \operatorname{cosec}^n \theta \cot \theta - n \sin^{n-1} \theta \cos \theta \times \frac{\sin \theta}{\sin \theta}$$

$$\frac{dy}{d\theta} = -n \operatorname{cosec}^n \theta \cot \theta - n \sin^n \theta \cot \theta$$

$$\frac{dy}{d\theta} = -n \cot \theta (\operatorname{cosec}^n \theta + \sin^n \theta)$$

$$\frac{dy}{dx} = \frac{-n \cot \theta (\operatorname{cosec}^n \theta + \sin^n \theta)}{-\cot \theta (\operatorname{cosec} \theta + \sin \theta)}$$

$$\left(\frac{dy}{dx} \right)^2 = \frac{n^2 (\operatorname{cosec}^n \theta + \sin^n \theta)^2}{(\operatorname{cosec} \theta + \sin \theta)^2}$$

$$x = \operatorname{cosec} \theta - \sin \theta$$

$$x^2 = \operatorname{cosec}^2 \theta + \sin^2 \theta - 2 + 2 - 2$$

$$x^2 = (\operatorname{cosec} \theta + \sin \theta)^2 - 4$$

$$x^2 + 4 = (\operatorname{cosec} \theta + \sin \theta)^2$$

$$y = \operatorname{cosec}^n \theta - \sin^n \theta$$

$$y^2 = \operatorname{cosec}^{2n} \theta + \sin^{2n} \theta - 2 + 2 - 2$$

$$y^2 + 4 = (\operatorname{cosec}^n \theta + \sin^n \theta)^2$$

$$\left(\frac{dy}{dx} \right)^2 = n^2 \frac{(y^2 + 4)}{(x^2 + 4)}$$

$$(x^2 + 4) \left(\frac{dy}{dx} \right)^2 - n^2 (y^2 + 4) = 0$$

7. If a curve is represented parametrically by the equations

$$x = \sin\left(t + \frac{7\pi}{12}\right) + \sin\left(t - \frac{\pi}{12}\right) + \sin\left(t + \frac{3\pi}{12}\right), \quad y = \cos\left(t + \frac{7\pi}{12}\right) + \cos\left(t - \frac{\pi}{12}\right) + \cos\left(t + \frac{3\pi}{12}\right)$$
 then find the

$$\text{value of } \frac{d}{dt} \left(\frac{x}{y} - \frac{y}{x} \right) \text{ at } t = \frac{\pi}{8}.$$

यदि एक वक्र निम्न प्राचलिक समीकरणों से दर्शाया जाता है

$$x = \sin\left(t + \frac{7\pi}{12}\right) + \sin\left(t - \frac{\pi}{12}\right) + \sin\left(t + \frac{3\pi}{12}\right), \quad y = \cos\left(t + \frac{7\pi}{12}\right) + \cos\left(t - \frac{\pi}{12}\right) + \cos\left(t + \frac{3\pi}{12}\right) \text{ तो } \frac{d}{dt} \left(\frac{x}{y} - \frac{y}{x} \right)$$

का $t = \frac{\pi}{8}$ पर मान ज्ञात कीजिए।

Ans. 8

Sol. $x = \sin\left(t + \frac{7\pi}{12}\right) + \sin\left(t - \frac{\pi}{12}\right) + \sin\left(t + \frac{3\pi}{12}\right)$

$$x = 2 \sin\left(\frac{t + \frac{7\pi}{12} + t - \frac{\pi}{12}}{2}\right) \cos\left(\frac{t + \frac{7\pi}{12} - t + \frac{\pi}{12}}{2}\right) + \sin\left(t + \frac{\pi}{4}\right)$$

$$x = 2 \sin\left(t + \frac{\pi}{4}\right) \cos\left(\frac{\pi}{3}\right) + \sin\left(t + \frac{\pi}{4}\right)$$

$$x = 2 \sin\left(t + \frac{\pi}{4}\right)$$

$$y = \cos\left(t + \frac{7\pi}{12}\right) + \cos\left(t - \frac{\pi}{12}\right) + \cos\left(t + \frac{3\pi}{12}\right)$$

$$y = 2 \cos\left(\frac{t + \frac{7\pi}{12} + t - \frac{\pi}{12}}{2}\right) \cos\left(\frac{t + \frac{7\pi}{12} - t + \frac{\pi}{12}}{2}\right) + \cos\left(t + \frac{\pi}{4}\right)$$

$$y = 2 \cos\left(t + \frac{\pi}{4}\right)$$

$$\frac{x}{y} - \frac{y}{x} = \frac{x^2 - y^2}{xy}$$

$$= \frac{4 \left[\sin^2\left(t + \frac{\pi}{4}\right) - \cos^2\left(t + \frac{\pi}{4}\right) \right]}{4 \sin\left(t + \frac{\pi}{4}\right) \cos\left(t + \frac{\pi}{4}\right)}$$

$$= \frac{-4 \cos\left(\frac{\pi}{2} + 2t\right)}{2 \sin\left(\frac{\pi}{2} + 2t\right)}$$

$$= +2 \frac{\sin 2t}{\cos 2t}$$

$$\left(\frac{x}{y} - \frac{y}{x}\right) = +2 \tan 2t$$

$$\frac{d}{dt}\left(\frac{x}{y} - \frac{y}{x}\right) = 2 \sec^2(2t) \times 2$$

$$= 4 \sec^2(2t)$$

$$= 4 \sec^2\left(2 \times \frac{\pi}{8}\right)$$

$$= 8$$

8. If $x = \frac{1 + \ell n t}{t^2}$ and $y = \frac{3 + 2 \ell n t}{t}$. Show that $y \frac{dy}{dx} = 2x \left(\frac{dy}{dx}\right)^2 + 1$.

यदि $x = \frac{1 + \ell n t}{t^2}$ एवं $y = \frac{3 + 2 \ell n t}{t}$ तो दर्शाइये $y \frac{dy}{dx} = 2x \left(\frac{dy}{dx}\right)^2 + 1$

Ans.

Sol. $x = \frac{1 + \ln t}{t^2}$

$$\frac{dx}{dt} = \frac{\frac{1}{t} \times t^2 - (1 + \ln t)(2t)}{t^4}$$

$$= \frac{t - 2t - 2t \ln t}{t^4}$$

$$\frac{dx}{dt} = \frac{-(1 + 2 \ln t)}{t^3}$$

$$y = \frac{3 + 2 \ln t}{t}$$

$$\frac{dy}{dt} = \frac{\frac{2}{t} \cdot t - (3 + 2 \ln t)}{t^2}$$

$$\frac{dy}{dt} = \frac{-(1 + 2 \ln t)}{t^2}$$

$$\frac{dy}{dx} = \frac{-(1 + 2 \ln t)}{t^2} \times \frac{t^3}{-(1 + 2 \ln t)}$$

$$\frac{dy}{dx} = t$$

$$\left(\frac{dy}{dx} \right)^2 = t^2$$

$$t^2 = \frac{1 + \ln t}{x} \quad t = \frac{3 + 2 \ln t}{y}$$

$$\left(\frac{dy}{dx} \right)^2 = \frac{1 + \ln t}{x}$$

$$1 + 2x \left(\frac{dy}{dx} \right)^2 = 2 + 2 \ln t + 1$$

$$1 + 2x \left(\frac{dy}{dx} \right)^2 = 3 + 2 \ln t \Rightarrow 1 + 2x \left(\frac{dy}{dx} \right)^2 = ty$$

$$1 + 2x \left(\frac{dy}{dx} \right)^2 = \left(\frac{dy}{dx} \right) y$$

9. Differentiate $\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$ w. r. t. $\sqrt{1-x^4}$.

$\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}$ का $\sqrt{1-x^4}$ के सापेक्ष अवकलन कीजिये।

Ans. $\frac{1 + \sqrt{1-x^4}}{x^6}$

Sol. $y = \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}} \times \frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} + \sqrt{1-x^2}}$

$$y = \frac{1+x^2+1-x^2+2\sqrt{1-x^4}}{1+x^2-1+x^2}$$

$$y = \frac{2+2\sqrt{1-x^4}}{2x^2} = \frac{1+\sqrt{1-x^4}}{x^2}$$

$$A = \sqrt{1-x^4}$$

$$A^2 = 1-x^4$$

$$x^4 = 1-A^2$$

$$x^2 = \sqrt{1-A^2}$$

$$y = \frac{1+A}{\sqrt{1-A^2}}$$

$$\frac{dy}{dA} = \frac{1\sqrt{1-A^2} - (1+A) \frac{-2A}{2\sqrt{1-A^2}}}{(1-A^2)}$$

$$= \frac{1 - A^2 + A + A^2}{\sqrt{1 - A^2}(1 - A^2)} = \frac{1 + A}{(1 - A^2)^{3/2}}$$

$$\frac{dy}{dA} = \frac{1 + \sqrt{1 - x^4}}{(x^4)^{3/2}} = \frac{1 + \sqrt{1 - x^4}}{x^6}$$

10. Let $g(x)$ be a polynomial, of degree one and $f(x)$ be defined by $f(x) = \begin{cases} g(x), & x \leq 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x}, & x > 0 \end{cases}$. Find the continuous function $f(x)$ satisfying $f'(1) = f(-1)$

माना $g(x)$ ईकाई घात का एक बहुपद है तथा $f(x)$ परिभाषित है $f(x) = \begin{cases} g(x), & x \leq 0 \\ \left(\frac{1+x}{2+x}\right)^{1/x}, & x > 0 \end{cases}$ सतत फलन $f(x)$ निकालिए जो

$f'(1) = f(-1)$ को संतुष्ट करता है।

Ans. $f(x) = \begin{cases} -\frac{2}{3} \left[\frac{1}{6} + \ln \frac{3}{2} \right] x & \text{if } x \leq 0 \\ \left(\frac{1+x}{2+x} \right)^{1/x} & \text{if } x > 0 \end{cases}$

Sol. $f(x) = \begin{cases} g(x), & x \leq 0 \\ \left(\frac{1+x}{2+x} \right)^{1/x}, & x > 0 \end{cases}$

$$g(x) = ax + b$$

$$f(x) = \left(\frac{1+x}{2+x} \right)^{\frac{1}{x}}$$

$$f(0^-) = f(0^+)$$

$$b = \left(\frac{1}{2} \right)^{\frac{1}{0}} \Rightarrow b = 0$$

$$\ln(f(x)) = \frac{1}{x} \log \left(\frac{1+x}{2+x} \right)$$

$$\frac{1}{f(x)} f'(x) = \frac{-1}{x^2} \log\left(\frac{x+1}{x+2}\right) + \frac{1}{x} \left(\frac{x+2}{x+1}\right) \left(\frac{x+2-x-1}{(x+2)^2}\right)$$

$$f'(x) = \left(\frac{x+1}{x+2}\right)^{\frac{1}{x}} \left[\frac{-1}{x^2} \log\left(\frac{x+1}{x+2}\right) + \frac{1}{x(x+1)(x+2)} \right]$$

$$f'(1) = \left(\frac{2}{3}\right) \left[-\log \frac{2}{3} + \frac{1}{1.2.3} \right]$$

$$f'(1) = \frac{2}{3} \left[\frac{1}{6} - \log \frac{2}{3} \right]$$

$$f(-1) = g(-1) = -a + b$$

$$\frac{2}{3} \left[\frac{1}{6} + \log \frac{3}{2} \right] = -a + 0$$

$$a = \frac{-2}{3} \left[\frac{1}{6} + \log \frac{3}{2} \right]$$

11. If $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3 \cdot (x^3 - y^3)$, prove that $\frac{dy}{dx} = \frac{x^2}{y^2} \sqrt{\frac{1-y^6}{1-x^6}}$.

यदि $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3 \cdot (x^3 - y^3)$ तो सिद्ध कीजिए $\frac{dy}{dx} = \frac{x^2}{y^2} \sqrt{\frac{1-y^6}{1-x^6}}$.

Ans.

Sol. $\sqrt{1-x^6} + \sqrt{1-y^6} = a^3 (x^3 - y^3)$

$$\frac{-6x^5}{2\sqrt{1-x^6}} + \frac{-6y^5}{2\sqrt{1-y^6}} y' = 3a^3 x^2 - 3a^3 y^2 y'$$

$$y' \left(\frac{y^5}{\sqrt{1-y^6}} - a^3 y^2 \right) = - \left(\frac{x^5}{\sqrt{1-x^6}} + a^3 x^2 \right)$$

$$y' = \frac{\sqrt{1-y^6}}{\sqrt{1-x^6}} \left(\frac{-x^5 - a^3 x^2 \sqrt{1-x^6}}{y^5 - a^3 y^2 \sqrt{1-y^6}} \right)$$

$$y' = \frac{\sqrt{1-y^6}}{\sqrt{1-x^6}} \times \frac{-x^2}{y^2} \left(\frac{x^3 + a^3 \sqrt{1-x^6}}{y^3 - a^3 \sqrt{1-y^6}} \right)$$

$$y' = \frac{\sqrt{1-y^6}}{\sqrt{1-x^6}} \times \frac{x^2}{y^2} \left(\frac{a^3 \sqrt{1-x^6} + x^3}{a^3 \sqrt{1-y^6} - y^3} \right)$$

$$\sqrt{1-x^6} + \sqrt{1-y^6} = a^3 x^3 - a^3 y^3$$

12. If $y = x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots}}}$, prove that $\frac{dy}{dx} = \frac{1}{2 - \frac{x}{x + \frac{1}{x + \frac{1}{x + \dots}}}}$.

यदि $y = x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots}}}$, तो सिद्ध कीजिए $\frac{dy}{dx} = \frac{1}{2 - \frac{x}{x + \frac{1}{x + \frac{1}{x + \dots}}}}$.

Ans.

Sol. $y = x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots}}}$

$$y = x + \frac{1}{y}$$

$$y^2 = xy + 1$$

$$2yy' = y + xy'$$

$$y'(2y - x) = y$$

$$y' = \frac{y}{2y - x}$$

$$y' = \frac{1}{2 - \frac{x}{y}}$$

$$\frac{dy}{dx} = \frac{1}{2 - \frac{x}{x + \frac{1}{x + \frac{1}{x + \dots}}}}$$

13. Let $f(x) = x + \frac{1}{2x + \frac{1}{2x + \frac{1}{2x + \dots \infty}}}$. Compute the value of $f(100) \cdot f'(100)$.

माना $f(x) = x + \frac{1}{2x + \frac{1}{2x + \frac{1}{2x + \dots \infty}}}$ तो $f(100) \cdot f'(100)$ का मान ज्ञात कीजिए।

Ans. 100

Sol. $y = x + \frac{1}{2x + \frac{1}{2x + \frac{1}{2x + \dots \infty}}}$

$$y - x = \frac{1}{2x + \frac{1}{2x + \frac{1}{2x + \dots \infty}}}$$

$$y - x = \frac{1}{2x + y - x}$$

$$y - x = \frac{1}{x + y}$$

$$(x + y)(y - x) = 1$$

$$y^2 - x^2 = 1$$

$$2yy' - 2x = 0$$

$$\frac{dy}{dx} = \frac{x}{y}$$

$$y \frac{dy}{dx} = x$$

$$f(x)f'(x) = x$$

$$f(100)f'(100) = 100$$

14. Find the derivative with respect to x of the function: $(\log_{\cos x} \sin x)(\log_{\sin x} \cos x)^{-1} + \arcsin \frac{2x}{1+x^2}$ at $x = \frac{\pi}{4}$.

निम्नलिखित फलन का x के सापेक्ष अवकलज निकाले : $(\log_{\cos x} \sin x)(\log_{\sin x} \cos x)^{-1} + \arcsin \frac{2x}{1+x^2}$, $x = \frac{\pi}{4}$ पर

Ans. $\frac{32}{16+\pi^2} - \frac{8}{\ln 2}$

Sol. $(\log_{\cos x} \sin x)(\log_{\sin x} \cos x)^{-1} + \arcsin \frac{2x}{1+x^2} = y$

$$\frac{\log \sin x}{\log \cos x} \times \frac{\log \sin x}{\log \cos x} + \sin^{-1} \left(\frac{2x}{1+x^2} \right) = y$$

$$y = \left(\frac{\ln \sin x}{\ln \cos x} \right)^2 + \tan^{-1} x$$

$$\frac{dy}{dx} = 2 \left(\frac{\ln \sin x}{\ln \cos x} \right) \left(\frac{\cot x \ln \cos x + \tan x \ln \sin x}{(\ln \cos x)^2} \right) + 2 \times \frac{1}{1+x^2}$$

$$x = \frac{\pi}{4}$$

$$\frac{dy}{dx} = \frac{32}{16+\pi^2} - \frac{8}{\ln 2}$$

15. Suppose $f(x) = \tan(\sin^{-1}(2x))$
- Find the domain and range of f .
 - Express $f(x)$ as an algebraic function of x .
 - Find $f'(1/4)$.

माना कि $f(x) = \tan(\sin^{-1}(2x))$

- फलन f का प्रांत तथा परिसर बताइए।
- विवक्षित x के बीजगणितीय फलन के रूप में प्रदर्शित करें।
- $f'(1/4)$ ज्ञात कीजिए।

Ans. (a) $\left(-\frac{1}{2}, \frac{1}{2}\right), (-\infty, \infty)$ (b) $f(x) = \frac{2x}{\sqrt{1-4x^2}}$ (c) $\frac{16\sqrt{3}}{9}$

Sol. $f(x) = \tan(\sin^{-1}(2x))$

(a) $-1 \leq 2x \leq 1$

$$\frac{-1}{2} \leq x \leq \frac{1}{2}$$

$$x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$$

$$x = \frac{-1}{2} \Rightarrow \tan\left(\frac{-\pi}{2}\right) \text{ Rejected}$$

$$x = \frac{1}{2} \Rightarrow \tan\left(\frac{\pi}{2}\right) \text{ Rejected}$$

Domain: $x \in \left(\frac{-1}{2}, \frac{1}{2}\right)$

$$f(x) = \tan\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$$

$$= (-\infty, \infty)$$

(b) $f(x) = \tan(\sin^{-1}(2x))$

$$= \tan\left(\tan^{-1} \frac{2x}{\sqrt{1-4x^2}}\right)$$

$$f(x) = \frac{2x}{\sqrt{1-4x^2}}$$

$$(c) \quad f'(x) = \frac{2\sqrt{1-4x^2} - 2x \left(\frac{-8x}{2\sqrt{1-4x^2}} \right)}{1-4x^2}$$

$$\Rightarrow \frac{2(1-4x^2) + 8x^2}{(1-4x^2)^{\frac{3}{2}}}$$

$$f'(x) = \frac{2}{(1-4x^2)^{\frac{3}{2}}}$$

$$x = \frac{1}{4}$$

$$f'\left(\frac{1}{4}\right) = \frac{16\sqrt{3}}{9}$$

16. If $y = \tan^{-1} \frac{1}{x^2+x+1} + \tan^{-1} \frac{1}{x^2+3x+3} + \tan^{-1} \frac{1}{x^2+5x+7} + \tan^{-1} \frac{1}{x^2+7x+13} + \dots$ to n terms. Find $\frac{dy}{dx}$, expressing your answer in 2 terms.

यदि $y = \tan^{-1} \frac{1}{x^2+x+1} + \tan^{-1} \frac{1}{x^2+3x+3} + \tan^{-1} \frac{1}{x^2+5x+7} + \tan^{-1} \frac{1}{x^2+7x+13} + \dots$ पदों तक $\frac{dy}{dx}$ निकालिए, अपना उत्तर दो पदों में दीजिये।

Ans. $\frac{1}{1+(x+n)^2} - \frac{1}{1+x^2}$

Sol. $y = \tan^{-1} \frac{1}{x^2+x+1} + \tan^{-1} \frac{1}{x^2+3x+3} + \tan^{-1} \frac{1}{x^2+5x+7} + \tan^{-1} \frac{1}{x^2+7x+13} + \dots$ to n terms

$$T_r = \tan^{-1} \left(\frac{1}{x^2 + (2r-1)x + (r^2 - r + 1)} \right)$$

$$T_r = \tan^{-1} \left(\frac{1}{1+x^2 + 2rx + r^2 - x - r} \right)$$

$$= \tan^{-1} \left(\frac{1}{1 + (x+r)^2 - (x+r)} \right)$$

$$= \tan^{-1} \left(\frac{1}{1 + (x+r)(x+r-1)} \right)$$

$$T_r = \tan^{-1}(x+r) - \tan^{-1}(x+r-1)$$

$$T_1 = \tan^{-1}(x+1) - \tan^{-1}(x)$$

$$T_2 = \tan^{-1}(x+2) - \tan^{-1}(x+1)$$

$$T_3 = \tan^{-1}(x+3) - \tan^{-1}(x+2)$$

$$\begin{array}{c} \cdot \\ \cdot \\ \cdot \end{array}$$

$$T_n = \tan^{-1}(x+n) - \tan^{-1}(x+n-1)$$

$$S_n = y = \tan^{-1}(x+n) - \tan^{-1}x$$

$$\frac{dy}{dx} = \frac{1}{1+(x+n)^2} - \frac{1}{1+x^2}$$

17. If $y = \tan^{-1} \frac{u}{\sqrt{1-u^2}}$ and $x = \sec^{-1} \frac{1}{2u^2-1}$, $u \in \left(0, \frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$ prove that $2 \frac{dy}{dx} + 1 = 0$.

यदि $y = \tan^{-1} \frac{u}{\sqrt{1-u^2}}$ तथा $x = \sec^{-1} \frac{1}{2u^2-1}$, $u \in \left(0, \frac{1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, 1\right)$ सिद्ध कीजिए $2 \frac{dy}{dx} + 1 = 0$

Ans.

Sol. $y = \tan^{-1} \frac{u}{\sqrt{1-u^2}}$

$$x = \sec^{-1} \frac{1}{2u^2-1}$$

$$u = \sin \theta$$

$$u = \cos \theta$$

$$y = \tan^{-1} \left(\frac{\sin \theta}{\cos \theta} \right)$$

$$x = \sec^{-1} \left(\frac{1}{2\cos^2 \theta - 1} \right)$$

$$x = \sec^{-1} \left(\frac{1}{\cos 2\theta} \right)$$

$$x = \sec^{-1}(\sec 2\theta)$$

$$y = \theta$$

$$x = 2\theta$$

$$y = \sin^{-1} u$$

$$x = 2 \cos^{-1} u$$

$$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}}$$

$$\frac{dx}{du} = \frac{-2}{\sqrt{1-u^2}}$$

$$\frac{dy}{dx} = \frac{-1}{2}$$

$$\frac{2dy}{dx} + 1 = 0$$

18. If $y = \cot^{-1} \frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}$, find $\frac{dy}{dx}$ if $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$.

यदि $y = \cot^{-1} \frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}$, $\frac{dy}{dx}$ निकालिये जब $x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$ हो।

Ans. $\frac{1}{2}$ or $-\frac{1}{2}$

Sol. $y = \cot^{-1} \frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}$

$$y = \cot^{-1} \left(\frac{1 + \sin x + 1 - \sin x + 2\sqrt{1 - \sin^2 x}}{1 + \sin x - 1 + \sin x} \right)$$

$$y = \cot^{-1} \left(\frac{1 + |\cos x|}{\sin x} \right)$$

$$x \in \left(0, \frac{\pi}{2}\right)$$

$$y = \cot^{-1} \left(\frac{1 + \cos x}{\sin x} \right)$$

$$y = \cot^{-1} \left(\cot \frac{x}{2} \right)$$

$$y = \frac{x}{2}$$

$$\frac{dy}{dx} = \frac{1}{2}$$

$$x \in \left(\frac{\pi}{2}, \pi\right)$$

$$y = \cot^{-1} \left(\frac{1 - \cos x}{\sin x} \right)$$

$$y = \cot^{-1} \left(\tan \frac{x}{2} \right)$$

$$y = \frac{\pi}{2} - \frac{x}{2}$$

$$\frac{dy}{dx} = -\frac{1}{2}$$

19. If $y = \tan^{-1} \frac{x}{1 + \sqrt{1-x^2}} + \sin \left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right)$, then find $\frac{dy}{dx}$ for $x \in (-1, 1)$.

यदि $y = \tan^{-1} \frac{x}{1 + \sqrt{1-x^2}} + \sin \left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right)$, तो $x \in (-1, 1)$ के लिये $\frac{dy}{dx}$ बताइये।

Ans. $\frac{1-2x}{2\sqrt{1-x^2}}$

Sol. $y = \tan^{-1} \frac{x}{1 + \sqrt{1-x^2}} + \sin \left(2 \tan^{-1} \sqrt{\frac{1-x}{1+x}} \right)$

$$x = \sin \theta \quad x = \cos \phi$$

$$y = \tan^{-1} \left(\frac{\sin \theta}{1 + \cos \theta} \right) + \sin \left(2 \tan^{-1} \sqrt{\frac{1 - \cos \phi}{1 + \sin \phi}} \right)$$

$$y = \tan^{-1} \left(\tan \frac{\theta}{2} \right) + \sin \left(2 \tan^{-1} \left(\tan \frac{\phi}{2} \right) \right)$$

$$y = \frac{\theta}{2} + \sin \left(2 \times \frac{\phi}{2} \right)$$

$$y = \frac{\theta}{2} + \sin(\phi)$$

$$y = \frac{\sin^{-1} x}{2} + \frac{\sqrt{1-x^2}}{1}$$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{1-x^2}} + \frac{-2x}{2\sqrt{1-x^2}}$$

$$\frac{dy}{dx} = \frac{1-2x}{2\sqrt{1-x^2}}$$

20. Let $P(x)$ be a polynomial of degree 4 such that $P(1) = P(3) = P(5) = P'(7) = 0$. If the real number $x \neq 1, 3, 5$ is such that $P(x) = 0$ can be expressed as $x = p/q$ where 'p' and 'q' are relatively prime, then $(p + q)$ equals.

माना $P(x)$ एक चार घात का बहुपद है जिसके लिये $P(1) = P(3) = P(5) = P'(7) = 0$ हो। यदि वास्तविक संख्या $x \neq 1, 3, 5$ इस प्रकार है कि $P(x) = 0$ को $x = p/q$ के रूप में प्रदर्शित किया जा सके, जहाँ 'p' तथा 'q' सापेक्षतः अभाज्य संख्याएँ (relatively prime) हैं, तो $(p + q)$ किसके बराबर होगा?

Ans. 100

Sol. $P(1) = P(3) = P(5) = P'(7) = 0$

$$P(x) = (x-1)(x-3)(x-5)(ax+b)$$

$$\log(p(x)) = \log(x-1) + \log(x-3) + \log(x-5) + \log(ax+b)$$

$$\frac{1}{P(x)} P'(x) = \frac{1}{x-1} + \frac{1}{x-3} + \frac{1}{x-5} + \frac{a}{ax+b}$$

$$x = 7$$

$$0 = \frac{1}{6} + \frac{1}{4} + \frac{1}{2} + \frac{a}{7a+b}$$

$$\frac{a}{7a+b} = \frac{-11}{12}$$

$$89a = -11b$$

$$\frac{-b}{a} = \frac{89}{11}$$

$$x = \frac{89}{11} = \frac{p}{q}$$

$$p + q = 100$$

$$21. \quad f(x) = \begin{cases} b \sin^{-1}\left(\frac{x+c}{2}\right), & -\frac{1}{2} < x < 0 \\ \frac{1}{2} & \text{at } x = 0 \\ \frac{e^{ax/2} - 1}{x}, & 0 < x < \frac{1}{2} \end{cases}$$

If $f(x)$ is differentiable at $x = 0$ and $|c| < 1/2$ then find the value of 'a' and prove that $64b^2 = 4 - c^2$.

$$f(x) = \begin{cases} b \sin^{-1}\left(\frac{x+c}{2}\right), & -\frac{1}{2} < x < 0 \\ \frac{1}{2} & \text{at } x = 0 \\ \frac{e^{ax/2} - 1}{x}, & 0 < x < \frac{1}{2} \end{cases}$$

यदि $f(x)$, $x = 0$ पर अवकलनीय है तथा $|c| < 1/2$ हो तो 'a' का मान बताइए एवं सिद्ध कीजिये कि $64b^2 = 4 - c^2$

Ans. $a = 1$

$$\text{Sol.} \quad f(x) = \begin{cases} b \sin^{-1}\left(\frac{x+c}{2}\right), & -\frac{1}{2} < x < 0 \\ \frac{1}{2} & \text{at } x = 0 \\ \frac{e^{ax/2} - 1}{x}, & 0 < x < \frac{1}{2} \end{cases}$$

$$f(0^-) = f(0) = f(0^+)$$

$$b \sin^{-1}\left(\frac{c}{2}\right) = \frac{1}{2} = \lim_{x \rightarrow 0^+} \frac{e^{\frac{ax}{2}} - 1}{x}$$

$$= \lim_{x \rightarrow 0^+} \frac{\frac{a}{2} e^{\frac{ax}{2}}}{1} = \frac{a}{2}$$

$$a = 1$$

$$f'(x) = \begin{cases} \frac{b\left(\frac{1}{2}\right)}{\sqrt{1 - \left(\frac{x+c}{2}\right)^2}} & ; -\frac{1}{2} < x < 0 \\ 0 & ; x = 0 \\ \frac{\frac{x}{e^2} \times x - \left(e^{\frac{x}{2}} - 1\right)}{x^2} & ; 0 < x < \frac{1}{2} \end{cases}$$

$$\text{LHD} = \text{RHD at } x = 0$$

$$\text{LHD} = \frac{b}{2\sqrt{1 - \frac{c^2}{4}}} = \frac{b}{\sqrt{4 - c^2}}$$

$$\text{RHD} = \lim_{x \rightarrow 0} \frac{\frac{x}{e^2} - e^{\frac{x}{2}} + 1}{x^2} = \lim_{x \rightarrow 0} \frac{e^{\frac{x}{2}}(x-2) + 2}{2x^2}$$

$$= \lim_{x \rightarrow 0} \frac{e^{\frac{x}{2}} + \frac{e^{\frac{x}{2}}(x-2)}{2}}{4x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{e^{\frac{x}{2}}}{2} + \frac{e^{\frac{x}{2}}(x-2)}{4} + \frac{e^{\frac{x}{2}}}{2}}{4}$$

$$= \frac{\frac{1}{2} + \frac{-2}{4} + \frac{1}{2}}{4}$$

$$= \frac{1}{8}$$

$$\text{RHD} = \text{LHD}$$

$$\frac{1}{8} = \frac{b}{\sqrt{4-c^2}}$$

$$\frac{1}{64} = \frac{b^2}{4-c^2}$$

$$64b^2 = 4 - c^2$$

22. If $f(x-y) = f(x) \cdot g(y) - f(y) \cdot g(x)$ and $g(x-y) = g(x) \cdot g(y) + f(x) \cdot f(y)$ for all $x, y \in \mathbb{R}$. If right hand derivative at $x=0$ exists for $f(x)$. Find derivative of $g(x)$ at $x=0$.

यदि $f(x-y) = f(x) \cdot g(y) - f(y) \cdot g(x)$ तथा $g(x-y) = g(x) \cdot g(y) + f(x) \cdot f(y)$ सभी $x, y \in \mathbb{R}$ है। यदि $x=0$ पर $f(x)$ का दायां अवकलज अस्तित्व में है। $x=0$ पर $g(x)$ का अवकलज बताइये।

Ans. $g'(0) = 0$

Sol. $f(x-y) = f(x) \cdot g(y) - f(y) \cdot g(x) \dots\dots\dots(1)$

$$g(x-y) = g(x) \cdot g(y) + f(x) \cdot f(y) \dots\dots\dots(2)$$

in (1) $x=y$

$$f(0) = f(x)g(x) - f(x)g(x)$$

$$f(0) = 0$$

$y=0$ in (1)

$$f(x) = f(x)g(0) - f(0)g(x)$$

$$f(x) = f(x)g(0) - (0)g(x)$$

$$g(0) = 1$$

$$f'(0^+) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(0)g(-h) - f(-h)g(0) - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(-h)}{-h} = f'(0^-)$$

$x=y$ in (2)

$$g(0) = g^2(x) + f^2(x)$$

$$0 = 2g(x)g'(x) + 2f(x)f'(x)$$

$$g'(0)g(0) + f(0)f'(0) = 0$$

$$g'(0) \times 1 + 0 = 0$$

23. Let $f(\theta) = \sin \left(\tan^{-1} \left(\frac{\sin \theta}{\sqrt{\cos 2\theta}} \right) \right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. Then the value of $\frac{d}{d(\tan \theta)}(f(\theta))$ [JEE-2011]

माना $f(\theta) = \sin \left(\tan^{-1} \left(\frac{\sin \theta}{\sqrt{\cos 2\theta}} \right) \right)$, जहाँ $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. तो $\frac{d}{d(\tan \theta)}(f(\theta))$ का मान होगा। [JEE-2011]

Ans. 1

Sol. Let $f(\theta) = \sin \alpha$ where $\alpha = \tan^{-1} \left(\frac{\sin \theta}{\sqrt{\cos 2\theta}} \right)$

$$\Rightarrow \tan \alpha = \frac{\sin \theta}{\sqrt{\cos 2\theta}}$$

$$\Rightarrow \sin \alpha = \frac{\sin \theta}{\cos \theta} = \tan \theta \quad \left(\because \theta \in \left(-\frac{\pi}{4}, \frac{\pi}{4} \right) \right)$$

$$\Rightarrow f(\theta) = \tan \theta$$

$$\Rightarrow \frac{d(f(\theta))}{d(\tan \theta)} = 1$$

DERIVATIVE OF INVERSE OF A FUNCTION

24. (a) Let $f(x) = x^2 - 4x - 3$, $x > 2$ and let g be the inverse of f . Find the value of g' where $f(x) = 2$.

(b) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$ and $g(x) = f^{-1}(x)$, then find $g'(-1)$.

(c) Suppose f^{-1} is the inverse function of a differentiable function f and let $G(x) = \frac{1}{f^{-1}(x)}$.

If $f(3) = 2$ and $f'(3) = \frac{1}{9}$, find $G'(2)$.

∴ द्धमाना कि $f(x) = x^2 - 4x - 3$, $x > 2$ तथा g, f का प्रतिलोम फलन है। g' का मान जबकि $f(x) = 2$ है क्या होगा?

(b) माना $f: \mathbb{R} \rightarrow \mathbb{R}$ में परिभाषित है $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$ तथा $g(x) = f^{-1}(x)$, हो तो $g'(-1)$ निकालिये।

(c) माना f^{-1} एक अवकलनीय फलन f का प्रतिलोम फलन है तथा $G(x) = \frac{1}{f^{-1}(x)}$ है।

यदि $f(3) = 2$ एवं $f'(3) = \frac{1}{9}$ तो $G'(2)$ का मान ज्ञात कीजिये।

Ans. (a) $\frac{1}{6}$ (b) $\frac{1}{14}$ (c) -1

Sol.

(a) $f(x) = x^2 - 4x - 3$ and $f(x) = 2$

For $x = 5, -1$

$$\Rightarrow x = 5 \quad (x > 2)$$

$$\text{Now } y = f(x) \Rightarrow f^{-1}(y) = x$$

$$\Rightarrow g(y) = x$$

$$\Rightarrow g'(y) = \frac{dx}{dy} = \frac{1}{2x-4} = \frac{1}{6}$$

(b) $f(x) = x^3 + 3x^2 + 6x - 5 + 4e^{2x}$

$$x = 0$$

$$f(0) = -1$$

$$g(f(x)) = x$$

$$g'(f(x)) \cdot f'(x) = 1$$

$$g'(f(x)) = \frac{1}{f'(x)}$$

$$x = 0$$

$$g'(f(0)) = \frac{1}{f'(0)}$$

$$f'(x) = 3x^2 + 6x + 6 + 8e^{2x}$$

$$f'(0) = 14$$

$$g'(-1) = \frac{1}{14}$$

(c) $G(x) = \frac{1}{f^{-1}(x)}$ $f(3) = 2$ $f'(3) = \frac{1}{9}$

$$h(x) = f^{-1}(x)$$

$$G(x) = \frac{1}{h(x)}$$

$$G'(x) = \frac{-h'(x)}{h^2(x)}$$

$$h(f(x)) = x$$

$$x = 3$$

$$h(2) = 3$$

$$h'(f(x))f'(x) = 1$$

$$h'(f(x)) = \frac{1}{f'(x)}$$

$$x = 3$$

$$h'(f(3)) = \frac{1}{f'(3)} \Rightarrow h'(2) = 9$$

$$G'(2) = \frac{-h'(2)}{[h(2)]^2} = \frac{-9}{(3)^2} = -1$$

25. If the function $f(x) = x^3 + e^{\frac{x}{2}}$ and $g(x) = f^{-1}(x)$, then the value of $g'(1)$ is.

[JEE-2009]

यदि फलन $f(x) = x^3 + e^{\frac{x}{2}}$ तथा $g(x) = f^{-1}(x)$ है, तो $g'(1)$ का मान ज्ञात कीजिए।

[JEE-2009]

Ans. 2

Sol. $f(x) = x^3 + e^{x/2}$, $g(x) = f^{-1}(x)$

$$\Rightarrow g'(f(x)) \cdot f'(x) = 1$$

$$\text{Put } f(x) = 1 \Rightarrow x^3 + e^{x/2} = 1$$

$$\Rightarrow x = 0$$

$$\Rightarrow g'(1) \cdot f'(0) = 1, f'(x) = 3x^2 + e^{x/2} \cdot \frac{1}{2}$$

$$\Rightarrow g'(1) = 2$$

HIGHER ORDER DERIVATIVE

26. (a) If $y = y(x)$ and it follows the relation $e^{xy} + y \cos x = 2$, then find (i) $y'(0)$ and (ii) $y''(0)$.

(b) A twice differentiable function $f(x)$ is defined for all real numbers and satisfies the following conditions

$$f(0) = 2; f'(0) = -5 \text{ and } f''(0) = 3.$$

The function $g(x)$ is defined by $g(x) = e^{ax} + f(x) \forall x \in \mathbb{R}$, where 'a' is any constant. If $g'(0) + g''(0) = 0$.

Find the value(s) of 'a'.

∴ यदि $y = y(x)$ है एवं यह सम्बन्ध $e^{xy} + y \cos x = 2$ के अनुगामी है तो (i) $y'(0)$ तथा (ii) $y''(0)$ निकालिये।

(b) यदि $f(x)$ एक दो बार अवकलनीय फलन है तथा वो निम्न शर्तों को संतुष्ट कर रहा है

$$f(0) = 2; f'(0) = -5 \text{ तथा } f''(0) = 3$$

फलन $g(x)$ परिभाषित है $g(x) = e^{ax} + f(x) \forall x \in \mathbb{R}$, जहाँ 'a' कोई नियतांक है। यदि $g'(0) + g''(0) = 0$ हो तो 'a' के मान बताइयें।

Ans. (a) (i) $y'(0) = -1$; (ii) $y''(0) = 2$ (b) $a = 1, -2$

Sol. (a) $y = y(x)$ $e^{xy} + y \cos x = 2$

$$x = 0$$

$$e^{0(y)} + y \cos(0) = 2$$

$$1 + y = 2$$

$$y(0) = 1$$

$$e^{xy} + y \cos x = 2$$

$$e^{xy}(y + xy') + y' \cos x - y \sin x = 0$$

$$x = 0$$

$$e^0(y(0) + 0 \times y') + y'(0) \cos 0 - y \sin 0 = 0$$

$$y'(0) = -1$$

$$e^{xy}(y + xy')^2 + e^{xy}(y' + y' + xy'') + y'' \cos x - y' \sin x - y' \sin x - y \cos x = 0$$

$$x = 0$$

$$y''(0) = 2$$

$$f(0) = 2$$

$$f'(0) = -5$$

$$f''(0) = 3$$

$$g(x) = e^{ax} + f(x)$$

$$g'(x) = ae^{ax} + f'(x)$$

$$g'(0) = a + f'(0)$$

$$= a - 5$$

$$g''(x) = a^2e^{ax} + f''(x)$$

$$g''(0) = a^2e^0 + f''(0)$$

$$g''(0) = a^2 + 3$$

$$g'(0) + g''(0) = 0$$

$$a - 5 + a^2 + 3 = 0$$

$$a^2 + a - 2 = 0$$

$$a = 1, -2$$

27. If $x = 2\cos t - \cos 2t$ and $y = 2\sin t - \sin 2t$, find the value of (d^2y/dx^2) when $t = (\pi/2)$.

यदि $x = 2\cos t - \cos 2t$ तथा $y = 2\sin t - \sin 2t$ तो (d^2y/dx^2) का $t = (\pi/2)$ पर मान बताइए।

Ans. $\frac{-3}{2}$

Sol. $x = 2\cos t - \cos 2t$

$$\frac{dx}{dt} = -2\sin t + 2\sin 2t$$

$$y = 2\sin t - \sin 2t$$

$$\frac{dy}{dt} = 2\cos t - 2\cos 2t$$

$$\frac{dy}{dx} = \frac{2\cos t - 2\cos 2t}{-2\sin t + 2\sin 2t} = \frac{\cos t - \cos 2t}{\sin 2t - \sin t}$$

$$\frac{dy}{dx} = \tan\left(\frac{3t}{2}\right)$$

$$\frac{d^2y}{dx^2} = \frac{3}{2} \sec^2\left(\frac{3t}{2}\right) \times \frac{dt}{dx}$$

$$= \frac{3}{2} \sec^2\left(\frac{3t}{2}\right) \times \frac{1}{2(\sin 2t - \sin t)}$$

$$t = \frac{\pi}{2}$$

$$\frac{d^2y}{dx^2} = \frac{-3}{2}$$

28. Find the value of the expression $y^3 \frac{d^2y}{dx^2}$ on the ellipse $3x^2 + 4y^2 = 12$.

दीर्घवृत्त $3x^2 + 4y^2 = 12$ पर व्यंजक $y^3 \frac{d^2y}{dx^2}$ का मान बताइए।

Ans. $\frac{-9}{4}$

Sol. $3x^2 + 4y^2 = 12$

$$6x + 8yy' = 0$$

$$8yy' = -6x$$

$$4yy' = -3x$$

$$4(y')^2 + 4yy'' = -3$$

$$\frac{d^2y}{dx^2} = \frac{-3 - 4\left(\frac{dy}{dx}\right)^2}{4y}$$

$$y^3 \frac{d^2 y}{dx^2} = \frac{-\left[3 + 4\left(\frac{dy}{dx}\right)^2\right] y^3}{4y}$$

$$= \frac{-\left[3 + 4\left(\frac{-3x}{4y}\right)^2\right] y^2}{4}$$

$$= -\left[\frac{12y^2 + 9x^2}{16y^2}\right] \times y^2$$

$$= -\left[\frac{3(12 - 3x^2) + 9x^2}{16}\right]$$

$$= -\frac{9}{4}$$

29. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a function such that $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$ for all $x \in \mathbb{R}$, then prove that $f(2) = f(1) - f(0)$.

यदि $f : \mathbb{R} \rightarrow \mathbb{R}$ एक फलन इस प्रकार है कि $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$ सभी $x \in \mathbb{R}$ के लिये हो तो सिद्ध कीजिये $f(2) = f(1) - f(0)$

Ans.

Sol. $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$

$$f'(1) = a \quad f''(2) = b$$

$$f'''(3) = c$$

$$f(x) = x^3 + ax^2 + bx + c$$

$$f'(x) = 3x^2 + 2ax + b$$

$$x = 1$$

$$a = 3 + 2a + b$$

$$a + b = -3 \dots\dots\dots (1)$$

$$f''(x) = 6x + 2a$$

$$x = 2$$

$$f''(2) = 12 + 2a$$

$$b = 12 + 2a \dots\dots\dots (2)$$

from (1) and (2)

$$a = -5$$

$$b = 2$$

$$f'''(x) = 6$$

$$x = 3$$

$$c = 6$$

$$f(x) = x^3 - 5x^2 + 2x + 6$$

$$f(2) = 8 - 20 + 4 + 6 = -2$$

$$f(1) = 4$$

$$f(0) = 6$$

$$f(2) = f(1) - f(0)$$

$$-2 = 4 - 6$$

$$-2 = -2$$

DIFFERENTIATION OF DETERMINANT

30. If $f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$ then $f'(x) = \lambda \cdot \begin{vmatrix} (x-a)^4 & (x-a)^2 & 1 \\ (x-b)^4 & (x-b)^2 & 1 \\ (x-c)^4 & (x-c)^2 & 1 \end{vmatrix}$. Find the value of λ .

यदि $f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$ तो $f'(x) = \lambda \cdot \begin{vmatrix} (x-a)^4 & (x-a)^2 & 1 \\ (x-b)^4 & (x-b)^2 & 1 \\ (x-c)^4 & (x-c)^2 & 1 \end{vmatrix}$ है। λ का मान ज्ञात कीजिए।

Ans. 3

Sol. $f(x) = \begin{vmatrix} (x-a)^4 & (x-a)^3 & 1 \\ (x-b)^4 & (x-b)^3 & 1 \\ (x-c)^4 & (x-c)^3 & 1 \end{vmatrix}$

$$f'(x) = \begin{vmatrix} 4(x-a)^3 & (x-a)^3 & 1 \\ 4(x-b)^3 & (x-b)^3 & 1 \\ 4(x-c)^3 & (x-c)^3 & 1 \end{vmatrix} + \begin{vmatrix} (x-a)^4 & 3(x-a)^2 & 1 \\ (x-b)^4 & 3(x-b)^2 & 1 \\ (x-c)^4 & 3(x-c)^2 & 1 \end{vmatrix} + \begin{vmatrix} (x-a)^4 & (x-a)^3 & 0 \\ (x-b)^4 & (x-b)^3 & 0 \\ (x-c)^4 & (x-c)^3 & 0 \end{vmatrix}$$

$$f'(x) = 0 + 3 \begin{vmatrix} (x-a)^4 & (x-a)^2 & 1 \\ (x-b)^4 & (x-b)^2 & 1 \\ (x-c)^4 & (x-c)^2 & 1 \end{vmatrix} + 0$$

$$\lambda = 3$$

31. If $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$ then find $f'(x)$.

यदि $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$ तो $f'(x)$ ज्ञात कीजिए।

Ans. $2(1+2x) \cdot \cos 2(x+x^2)$

Sol. $f(x) = \begin{vmatrix} \cos(x+x^2) & \sin(x+x^2) & -\cos(x+x^2) \\ \sin(x-x^2) & \cos(x-x^2) & \sin(x-x^2) \\ \sin 2x & 0 & \sin 2x^2 \end{vmatrix}$

$$f(x) = \sin 2x [\cos(x^2 + x - x + x^2)] + \sin 2x^2 [\cos(x + x^2 + x - x^2)]$$

$$= \sin 2x \cos 2x^2 + \sin 2x^2 \cos 2x$$

$$f(x) = \sin(2x^2 + 2x)$$

$$f'(x) = (4x + 2) \cos(2x^2 + 2x)$$

$$= 2(2x + 1) \cos 2(x + x^2)$$

32. If α be a repeated root of a quadratic equation $f(x) = 0$ and $A(x)$, $B(x)$, $C(x)$ be the polynomials of degree 3, 4

and 5 respectively, then show that $\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ is divisible by $f(x)$, where dash denotes the derivative.

यदि α द्विघात समीकरण $f(x) = 0$ का एक दोहराया गया मूल है तथा $A(x)$, $B(x)$, $C(x)$ क्रमशः 3, 4 तथा 5 घात के बहुपद हो तो

सिद्ध कीजिये $\begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$ फलन $f(x)$ द्वारा भाज्य है जहाँ y' का तात्पर्य अवकलज है।

Ans.

Sol. $f(x) = c(x - \alpha)^2$

$$F(x) = \begin{vmatrix} A(x) & B(x) & C(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$

Now $F(\alpha) = 0 \quad \therefore (x - \alpha)$ is factor of $F(x)$.

$$F'(x) = \begin{vmatrix} A'(x) & B'(x) & C'(x) \\ A(\alpha) & B(\alpha) & C(\alpha) \\ A'(\alpha) & B'(\alpha) & C'(\alpha) \end{vmatrix}$$

Now $F'(\alpha) = 0 \Rightarrow x - \alpha$ is a factor of $F'(x)$. So $(x - \alpha)$ must be repeated at least two times in $F(x)$.

$\Rightarrow F(x)$ is divisible by $f(x)$.

L'HOSPITAL RULE

Evaluate the following limits using L'Hospital's Rule or otherwise :

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33. $\lim_{x \rightarrow 0} \left[\frac{1}{x \sin^{-1} x} - \frac{1 - x^2}{x^2} \right]$

Ans. $\frac{5}{6}$

Sol. $\lim_{x \rightarrow 0} \left[\frac{1}{x \sin^{-1} x} - \frac{1 - x^2}{x^2} \right]$

$$x = \sin \theta$$

$$= \lim_{\theta \rightarrow 0} \left(\frac{1}{\theta \sin \theta} - \frac{1 - \sin^2 \theta}{\sin^2 \theta} \right)$$

$$= \lim_{\theta \rightarrow 0} \left(\frac{\sin \theta - \theta + \theta \sin^2 \theta}{\theta \sin^2 \theta} \right)$$

$$= \lim_{\theta \rightarrow 0} \left(\frac{\theta - \frac{\theta^3}{3!} + \dots - \theta + \theta^3 \left(\frac{\sin \theta}{\theta} \right)^2}{\theta^3 \left(\frac{\sin \theta}{\theta} \right)^2} \right)$$

$$= \lim_{\theta \rightarrow 0} \left(\frac{-\frac{1}{3!} + 1}{1} \right)$$

$$= \frac{5}{6}$$

34. $\lim_{x \rightarrow 0} \frac{x + \ln(\sqrt{x^2 + 1} - x)}{x^3}$

Ans. $\frac{1}{6}$

Sol. $\lim_{x \rightarrow 0} \frac{x + \ln(\sqrt{x^2 + 1} - x)}{x^3}$

$$\lim_{x \rightarrow 0} \frac{1 + \frac{1}{\sqrt{x^2 + 1} - x} \left(\frac{2x}{2\sqrt{x^2 + 1}} - 1 \right)}{3x^2}$$

$$\lim_{x \rightarrow 0} \frac{1 + \frac{1}{\sqrt{x^2 + 1} - x} \left(\frac{x - \sqrt{x^2 + 1}}{\sqrt{x^2 + 1}} \right)}{3x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \frac{1}{\sqrt{x^2 + 1}}}{3x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2} (x^2 + 1)^{-\frac{3}{2}} (2x)}{6x}$$

$$= \frac{1}{6} (1)^{-\frac{3}{2}} = \frac{1}{6}$$

35. $\lim_{x \rightarrow 0} \left[\frac{1}{x^2} - \frac{1}{\sin^2 x} \right]$

Ans. $-\frac{1}{3}$

Sol. $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$

$$\lim_{x \rightarrow 0} \frac{\sin^2 x - x^2}{x^2 \sin^2 x}$$

$$\lim_{x \rightarrow 0} \frac{(\sin x - x)(\sin x + x)}{x^2 \sin^2 x}$$

$$\lim_{x \rightarrow 0} \frac{\left(x - \frac{x^3}{3} + \dots - x \right) (x + \sin x)}{x^2 \sin^2 x}$$

$$\lim_{x \rightarrow 0} \frac{x^3 \left(-\frac{1}{6} + \dots \right) \times 2 \left(1 + \frac{\sin x}{x} \right)}{x^4 \left(\frac{\sin x}{x} \right)^2}$$

$$\frac{-\frac{1}{6} \times (1+1)}{1} = -\frac{1}{3}$$

36. (a) $\lim_{x \rightarrow 0^+} x^{(x^x - 1)}$

(b) $\lim_{x \rightarrow 0^+} (\cot x - \ln^2 x)$

Ans. (a) 1 (b) does not exist

Sol. (a) $\lim_{x \rightarrow 0^+} (x)^{(x^x - 1)}$

$$= (e)^{\lim_{x \rightarrow 0^+} (x^x - 1) \ln x} = (e)^2$$

$$L = \lim_{x \rightarrow 0^+} (x^x - 1) \ln x$$

$$= \lim_{x \rightarrow 0^+} \frac{x^x - 1}{\frac{1}{\ln x}}$$

Apply L' Hospital Rule

$$= \lim_{x \rightarrow 0^+} \frac{x^x (1 + \ln x)}{-\frac{1}{(\ln x)^2} \left(\frac{1}{x} \right)}$$

$$\lim_{x \rightarrow 0^+} -x (\ln x)^2 (1 + \ln x) \quad \left(\because \lim_{x \rightarrow 0} x^x = 1 \right)$$

put $\ln x = t$

$$\lim_{x \rightarrow -\infty} -e^t t^2 (1 + t)$$

put $t = -u$

$$\lim_{u \rightarrow \infty} -e^{-u} u^2 (1 - u)$$

$$L = \lim_{u \rightarrow \infty} \frac{-u^2 (1 - u)}{e^u} = 0$$

$$\Rightarrow e^u = e^0 = 1$$

$$(b) \lim_{x \rightarrow 0^+} (\cot x - \ln^2 x)$$

$$\lim_{x \rightarrow 0^+} \frac{\cos x - \sin x \ln^2 x}{\sin x}$$

DNE

$$37. \lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \ln(1 - x)}{x \cdot \tan^2 x}$$

Ans. $-\frac{1}{2}$

Sol. $\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \ln(1-x)}{x \cdot \tan^2 x}$

$$\lim_{x \rightarrow 0} \frac{1 + x - \frac{x^3}{3} + \dots - 1 + \frac{x^2}{2} - \frac{x^4}{4} \dots - x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} \dots}{x^3 \left(\frac{\tan x}{x} \right)^2}$$

$$\lim_{x \rightarrow 0} \frac{x^3 \left(-\frac{1}{6} + \dots - \frac{x}{24} - \frac{1}{3} \dots \right)}{(1)}$$

$$= -\frac{1}{6} - \frac{1}{3} = -\frac{1}{2}$$

38. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x - (\sin x)^{\sin x}}{1 - \sin x + \ln(\sin x)}$

Ans. 2

Sol. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x - (\sin x)^{\sin x}}{1 - \sin x + \ln(\sin x)}$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x - (\sin x)^{\sin x} [\cos x \log \sin x + \cos x]}{-\cos x + \frac{\cos x}{\sin x}}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - (\sin x)^{\sin x} [\log \sin x + 1]}{-1 + \frac{1}{\sin x}}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - (\sin x)^{\sin x} [\log(\sin x) + 1]}{\operatorname{cosec} x - 1}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{-(\sin x)^{\sin x} (\cos x \log \sin x + \cos x) - (\sin x)^{\sin x} \frac{\cos x}{\sin x}}{-\operatorname{cosec} x \cot x}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{-(\sin x)^{\sin x} \left[\log \sin x + 1 + \frac{1}{\sin x} \right]}{-\frac{1}{\sin^2 x}}$$

$$+ (1)^1 \frac{(0+1+1)}{+\frac{1}{1}} = 2$$

39. If $\lim_{x \rightarrow 0} \frac{a \sin x - bx + cx^2 + x^3}{2x^2 \cdot \ln(1+x) - 2x^3 + x^4}$ exists and is finite, find the values of a, b, c and the limit.

यदि सीमा $\lim_{x \rightarrow 0} \frac{a \sin x - bx + cx^2 + x^3}{2x^2 \cdot \ln(1+x) - 2x^3 + x^4}$ अस्तित्व में है एवं नियत हो तो a, b, c तथा सीमा के मान बताइये।

Ans. $a = 6, b = 6, c = 0; \frac{3}{40}$

Sol. $\lim_{x \rightarrow 0} \frac{a \sin x - bx + cx^2 + x^3}{2x^2 \ln(1+x) - 2x^3 + x^4}$

$$\lim_{x \rightarrow 0} \frac{ax - \frac{ax^3}{6} + \frac{ax^5}{120} + \dots - bx + cx^2 + x^3}{2x^2 \left(\frac{x}{1} - \frac{x^2}{2} + \frac{2x^5}{3} + \dots \right) - 2x^3 + x^4}$$

$$a - b = 0$$

$$a = b \quad c = 0 \quad -\frac{a}{6} + 1 = 0$$

$$a = 6$$

$$b = 6$$

$$\text{limit} = \frac{a}{120 \times \frac{2}{3}} = \frac{6}{80} = \frac{3}{40}$$

40. Evaluate : $\lim_{x \rightarrow 0} \frac{x^{6000} - (\sin x)^{6000}}{x^2 \cdot (\sin x)^{6000}}$

मान लिखिये : $\lim_{x \rightarrow 0} \frac{x^{6000} - (\sin x)^{6000}}{x^2 \cdot (\sin x)^{6000}}$

Ans. 1000

Sol. $\lim_{x \rightarrow 0} \frac{x^{6000} - (\sin x)^{6000}}{x^2 (\sin x)^{6000}}$

$$\lim_{x \rightarrow 0} \frac{\left(\frac{x}{\sin x}\right)^{6000} - 1}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{6000 \left(\frac{x}{\sin x}\right)^{5999} \left[\frac{\sin x - x \cos x}{\sin^2 x} \right]}{2x}$$

$$\lim_{x \rightarrow 0} \frac{6000 \left(\frac{x}{\sin x}\right)^{5999} \left[x - \frac{x^3}{3} + \dots - x + \frac{x^3}{2} \right]}{2x \frac{\sin^2 x}{x^2} \times x^2}$$

$$\lim_{x \rightarrow 0} \frac{6000 \left(\frac{x}{\sin x}\right)^{5999} \left(x^3 \left(\frac{1}{2} - \frac{1}{6} \right) \right)}{2x^3 \left(\frac{\sin x}{x} \right)^2}$$

$$\Rightarrow \frac{6000 \times (1) \left(\frac{2}{6} \right)}{2} = 1000$$

41. (a) If $\lim_{x \rightarrow 0} \frac{1 - \cos x \cdot \cos 2x \cdot \cos 3x \dots \cos nx}{x^2}$ has the value equal to 253, find the value of n (where n ∈ N).

(b) Find the value of $\lim_{x \rightarrow 0} \frac{1 - \cos^5 x \cos^3 2x \cos^3 3x}{x^2}$.

(c) If $\lim_{x \rightarrow 0} \frac{1 - \cos 3x \cdot \cos 9x \cdot \cos 27x \dots \cos 3^n x}{1 - \cos \frac{1}{3}x \cdot \cos \frac{1}{9}x \cdot \cos \frac{1}{27}x \dots \cos \frac{1}{3^n}x} = 3^{10}$, find the value of n.

(a) यदि $\lim_{x \rightarrow 0} \frac{1 - \cos x \cdot \cos 2x \cdot \cos 3x \dots \cos nx}{x^2}$ का मान 253 के बराबर है, तो n का मान बताइए (जहाँ $n \in \mathbb{N}$)

(b) $\lim_{x \rightarrow 0} \frac{1 - \cos^5 x \cos^3 2x \cos^3 3x}{x^2}$ का मान बताइये।

(c) यदि $\lim_{x \rightarrow 0} \frac{1 - \cos 3x \cdot \cos 9x \cdot \cos 27x \dots \cos 3^n x}{1 - \cos \frac{1}{3}x \cdot \cos \frac{1}{9}x \cdot \cos \frac{1}{27}x \dots \cos \frac{1}{3^n}x} = 3^{10}$ तो n का मान बताइए।

Ans. (a) 11, (b) 22, (c) 4

Sol. (a) $\lim_{x \rightarrow 0} \frac{1 - \cos x \cos 2x \cos 3x \dots \cos nx}{x^2}$

$$\lim_{x \rightarrow 0} \frac{1 - f(x)}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{-f'(x)}{2x}$$

$$f(x) = \cos x \cos 2x \cos 3x \dots \cos nx$$

$$\log [f(x)] = \log (\cos x) + \log (\cos 2x) + \log (\cos 3x) + \dots + \log (\cos nx)$$

$$\frac{1}{f(x)} f'(x) = -\frac{\sin x}{\cos x} - \frac{2 \sin 2x}{\cos 2x} - \frac{3 \sin 3x}{\cos 3x} \dots - \frac{n \sin nx}{\cos nx}$$

$$f'(x) = -(\cos x \cos 2x \cos 3x \dots \cos nx) (\tan x + 2 \tan 2x + \dots + n \tan nx)$$

$$\lim_{x \rightarrow 0} (\cos x \cos 2x \cos 3x \dots \cos nx) \frac{(\tan x + 2 \tan 2x + \dots + n \tan nx)}{2x}$$

$$= (1) \left(\frac{1}{2} \right) (1^2 + 2^2 + \dots + n^2) = 253$$

$$\frac{n(n+1)(2n+1)}{6} = 506$$

$$n(n+1)(2n+1) = 3036$$

$$n = 11$$

$$(b) \lim_{x \rightarrow 0} \frac{1 - \cos^5 x \cos^3 2x \cos^3 3x}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{1 - f(x)}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{-f'(x)}{2x}$$

$$f(x) = \cos^5 x \cos^3 2x \cos^3 3x$$

$$\log(f(x)) = 5 \log \cos x + 3 \log \cos 2x + 3 \log \cos 3x$$

$$\frac{1}{f(x)} f'(x) = 5 - \frac{\sin x}{\cos x} + 3 \times 2 \left(\frac{-\sin 2x}{\cos 2x} \right) + 3 \times 3 \left(\frac{-\sin 3x}{\cos 3x} \right)$$

$$f'(x) - f(x) [5 \tan x + 6 \tan 2x + 9 \tan 3x]$$

$$\Rightarrow \lim_{x \rightarrow 0} + \cos^5 x \cos^3 2x \cos^3 3x \frac{(5 \tan x + 6 \tan 2x + 9 \tan 3x)}{2x}$$

$$= (1) \left(\frac{1}{2} \right) (5 + 12 + 27)$$

$$= \frac{44}{2} = 22$$

$$(c) \lim_{x \rightarrow 0} \frac{1 - \cos 3x \cdot \cos 9x \cdot \cos 27x \dots \cos 3^n x}{1 - \cos \frac{x}{3} \cos \frac{x}{9} \cos \frac{x}{27} \dots \cos \frac{x}{3^n}}$$

$$\lim_{x \rightarrow 0} \frac{1 - f(x)}{1 - g(x)}$$

$$\lim_{x \rightarrow 0} \frac{+f'(x)}{+g'(x)}$$

$$f(x) = \cos 3x \cos 9x \cos 27x \dots \cos 3^n x$$

$$\log[f(x)] = \log(\cos 3x) + \log(\cos 9x) + \log(\cos 27x) + \dots + \log(\cos 3^n x)$$

$$\frac{f'(x)}{f(x)} = -\frac{3 \sin 3x}{\cos 3x} - 3^2 \frac{\sin 9x}{\cos 9x} - 3^3 \frac{\sin 27x}{\cos 27x} \dots - 3^n \frac{\sin 3^n x}{\cos 3^n x}$$

$$f'(x) = -f(x) [3 \tan 3x + 3^2 \tan 3^2 x + 3^3 \tan 3^3 x + \dots + 3^n \tan 3^n x]$$

$$g(x) = \cos \frac{x}{3} \cos \frac{x}{9} \cos \frac{x}{27} \dots \cos \frac{x}{3^n}$$

$$\log [g(x)] = \log \left(\cos \frac{x}{3} \right) + \log \left(\cos \frac{x}{9} \right) + \log \left(\cos \frac{x}{27} \right) + \dots + \log \left(\frac{x}{3^n} \right)$$

$$\frac{g'(x)}{g(x)} = -\frac{1}{3} \tan \frac{x}{3} - \frac{1}{9} \tan \frac{x}{9} - \frac{1}{27} \tan \frac{x}{27} \dots - \frac{1}{3^n} \tan \frac{x}{3^n}$$

$$g'(x) = g(x) \left[\frac{1}{3} \tan \frac{x}{3} + \frac{1}{9} \tan \frac{x}{9} + \frac{1}{27} \tan \frac{x}{27} + \dots + \frac{1}{3^n} \tan \frac{x}{3^n} \right]$$

$$= \lim_{x \rightarrow 0} \frac{+ \left(\cos 3x \cos 9x \dots \cos 3^n x \right) \left(3 \tan 3x + 3^2 \tan 3^2 x + 3^3 \tan 3^3 x \dots + 3^n \tan 3^n x \right)}{+ \left(\cos \frac{x}{3} \cos \frac{x}{9} \dots \cos \frac{x}{3^n} \right) \left(\frac{1}{3} \tan \frac{x}{3} + \frac{1}{3^2} \tan \frac{x}{9} + \dots + \frac{1}{3^n} \tan \frac{x}{3^n} \right)}$$

$$\frac{(1) \left(3^2 + 3^4 + 3^6 + \dots + 3^{2n} \right)}{(1) \left(\frac{1}{3^2} + \frac{1}{3^4} + \frac{1}{3^6} + \dots + \frac{1}{3^{2n}} \right)}$$

$$\Rightarrow \frac{3^2 \left(\frac{3^{2n} - 1}{3^2 - 1} \right)}{\frac{1}{3^2} \left(\frac{1 - \frac{1}{3^{2n}}}{1 - \frac{1}{3^2}} \right)} = \frac{9 \left(\frac{3^{2n} - 1}{8} \right)}{\frac{1}{9} \left(\frac{3^{2n} - \frac{1}{3^{2n}}}{\frac{3^2 - 1}{3^2}} \right)}$$

$$\Rightarrow 9 \cdot 3^{2n} = 3^{10}$$

$$3^{2n+2} = 3^{10}$$

$$2n + 2 = 10$$

$$2n = 8$$

$$n = 4$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 42 to 43

Let $f(x)$ and $g(x)$ be two differentiable functions, defined as : $f(x) = x^2 + xg'(1) + g''(2)$ and $g(x) = f(1)x^2 + xf'(x) + f''(x)$

माना $f(x)$ तथा $g(x)$ दो अवकलनीय फलन इस प्रकार परिभाषित हैं कि $f(x) = x^2 + xg'(1) + g''(2)$ तथा $g(x) = f(1)x^2 + xf'(x) + f''(x)$

42. The value of $f(1) + g(-1)$ is :

$f(1) + g(-1)$ का मान होगा :

- (A) 0 (B) 1 (C) 2 (D) 3

Ans. D

Sol. $f(x) = x^2 + xg'(1) + g''(2)$ $g(x) = f(1)x^2 + xf'(x) + f''(x)$

$$f(x) = x^2 + ax + b \qquad g(x) = (x^2 + xf'(x) + f''(x))$$

$$a = g'(1) \qquad b = g''(2) \qquad c = f(1)$$

$$\left. \begin{aligned} f'(x) &= 2x + a \\ f''(x) &= 2 \end{aligned} \right\} \Rightarrow$$

$$g(x) = cx^2 + x(2x + a) + 2$$

$$g(x) = (c + 2)x^2 + ax + 2$$

$$g'(x) = 2(c + 2)x + a$$

$$g'(1) = 2c + 4 + a = 0$$

$$\Rightarrow 2c + 4 = 0$$

$$c = -2$$

$$g''(x) = 2(c + 2)$$

$$g''(2) = 2(c + 2)$$

$$b = 2(c + 2) \Rightarrow b = 0$$

$$f(x) = x^2 + x(2c + a + 4) + 2(c + 2)$$

$$x = 1$$

$$f(1) = c = 1 + 2c + a + 4 + 2c + 4$$

$$-2 = 1 - 4 + a + 4 - 4 + 4$$

$$a = -3$$

$$f(x) = x^2 - 3x \qquad g(x) = -3x + 2$$

$$f(1) = 1 - 3$$

$$g(-1) = 3 + 2 = 5$$

$$= -2$$

$$f(1) + g(-1) = -2 + 5 = 3$$

43. The number of integers in the domain of the function $F(x) = \sqrt{-\frac{f(x)}{g(x)}} + \sqrt{3-x}$ is :

(A) 0

(B) 1

(C) 2

(D) Infinite

$F(x) = \sqrt{-\frac{f(x)}{g(x)}} + \sqrt{3-x}$ के प्रान्त में पूर्णांक मानों की संख्या होगी :

(A) 0

(B) 1

(C) 2

(D) अनन्त

Ans. C

Sol.
$$F(x) = \sqrt{\frac{-(x^2 - 3x)}{-(3x - 2)}} + \sqrt{3 - x}$$

$$= \sqrt{\frac{x^2 - 3x}{3x - 2}} + \sqrt{3 - x}$$

$$\frac{x^2 - 3x}{3x - 2} \geq 0$$

$$3 - x \geq 0$$

$$x - 3 \leq 0$$

$$x \leq 3$$

$$\frac{x(x-3)}{3x-2} \geq 0$$

$$x \in \left[0, \frac{2}{3}\right) \cup [3, \infty)$$

$$F.A. \Rightarrow x = \left[0, \frac{2}{3}\right) \cup \{3\}$$

$$I = 0, 3$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

44. Let $f(x) = \frac{\sqrt{x-2}\sqrt{x-1}}{\sqrt{x-1}-1}$ x. then :

(A) $f'(10) = 1$

(B) $f'(3/2) = -1$

(C) domain of $f(x)$ is $x \geq 1$

(D) None of these

माना $f(x) = \frac{\sqrt{x-2}\sqrt{x-1}}{\sqrt{x-1}-1}$. x तब :

(A) $f'(10) = 1$

(B) $f'(3/2) = -1$

(C) domain of $f(x)$ is $x \geq 1$

(D) इनमें से कोई नहीं

Ans. AB

Sol. $f(x) = \frac{\sqrt{(\sqrt{x-1})^2 + 1 - 2\sqrt{x-1}}}{\sqrt{x-1}-1} \cdot x = \frac{|\sqrt{x-1}-1|}{\sqrt{x-1}-1} \cdot x = \begin{cases} -x & \text{if } x \in [1, 2) \\ x & \text{if } x \in (2, \infty) \end{cases}$

45. If $f(x) = x \cdot |x|$, then its derivative is :

यदि $f(x) = x \cdot |x|$, तब $f(x)$ का अवकलन है :

(A) $2x$

(B) $-2x$

(C) $2|x|$

(D) $2x \operatorname{sgn} x$

Ans. CD

Sol. $f(x) = x \cdot |x|$

$x \geq 0$

$x < 0$

$f(x) = x^2$

$f(x) = -x^2$

$f'(x) = 2x$

$f'(x) = -2x$

$f'(x) = 2|x|$

46. Let $f(x) = \cos^{-1}(2x^2 - 1)$ then :

(A) $f(x)$ is continuous in $[-1, 1]$

(B) $f(x)$ is derivable in $(-1, 1)$

(C) range of $f(x)$ is $[0, \pi]$

(D) derivative of $f(x)$ w.r.t. $\sin^{-1}x$ at $x = 1/2$ is 2

माना $f(x) = \cos^{-1}(2x^2 - 1)$ तब :

(A) िगद्वय 1 ए 1, में सतत् है

(B) िगद्वय; 1 ए 1 द्वमें अवकलनीय है

(C) िगद्वका परिसर $[0, \pi]$ है

(D) िगद्वका $\sin^{-1}x$ के सापेक्ष $x = 1/2$ पर अवकलन 2 है

Ans. AC

Sol. $f(x) = \cos^{-1}(2x^2 - 1)$

$$-1 \leq 2x^2 - 1 \leq 1$$

$$0 \leq 2x^2 \leq 2$$

$$0 \leq x^2 \leq 1$$

$$x \in [-1, 1] \rightarrow \text{Continuous}$$

$$\text{Range} \Rightarrow [0, \pi]$$

$$\xi = \chi \circ \sigma \theta$$

$$f(x) = \cos^{-1}(2 \cos^2 \theta - 1)$$

$$= 2\theta$$

$$f(x) = \cos^{-1} x$$

$$f'(x) = \frac{-2}{\sqrt{1-x^2}}$$

$$g(x) = \sin^{-1} x$$

$$g'(x) = \frac{1}{\sqrt{1-x^2}}$$

$$= -2$$

MATCH THE COLUMN TYPE QUESTION(S)

47. Match the following column :

Column-I

$$(A) f(x) = \begin{cases} \ln(1+x^3) \cdot \sin \frac{1}{x}, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$

$$(B) g(x) = \begin{cases} \ln^2(1+x) \cdot \sin \frac{1}{x}, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$

$$(C) u(x) = \begin{cases} \ln \left(1 + \frac{\sin x}{2} \right), & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$

Column-II

(P) continuous everywhere but not differentiable at $x=0$

(Q) differentiable at $x=0$ but derivative is discontinuous at $x=0$

(R) differentiable and has continuous derivative

$$(D) v(x) = \lim_{t \rightarrow 0} \frac{2x}{\pi} \tan^{-1} \left(\frac{2}{t^2} \right)$$

(S) continuous and differentiable at $x = 0$

निम्न कॉलम को सुमेलित कीजिए :

Column-I

$$(A) f(x) = \begin{cases} \ln(1+x^3) \cdot \sin \frac{1}{x}, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$

$$(B) g(x) = \begin{cases} \ln^2(1+x) \cdot \sin \frac{1}{x}, & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$

$$(C) u(x) = \begin{cases} \ln \left(1 + \frac{\sin x}{2} \right), & \text{if } x > 0 \\ 0, & \text{if } x \leq 0 \end{cases}$$

$$(D) v(x) = \lim_{t \rightarrow 0} \frac{2x}{\pi} \tan^{-1} \left(\frac{2}{t^2} \right)$$

Column-II

(P) प्रत्येक बिन्दु पर संतत् परन्तु $x = 0$ पर अवकलनीय नहीं

(Q) $x > 0$ पर अवकलनीय परन्तु इसका अवकलज $x = 0$ पर असंतत् है

(R) अवकलनीय तथा अवकलज भी संतत् है

(S) $x > 0$ पर अवकलनीय तथा संतत् है

Ans. (A) R, S; (B) Q, S; (C) P; (D) R, S

Sol. (A) $f(x) = \begin{cases} \ln(1+x^3) \sin \frac{1}{x} & ; x > 0 \\ 0 & ; x \leq 0 \end{cases}$

$$f(0^+) = 0 \times [-1, 1] \Rightarrow 0$$

continuous

$$\text{LHD} = 0$$

$$\text{RHD} \Rightarrow \lim_{x \rightarrow 0} \frac{\ln(1+x^3) \sin \frac{1}{x} - 0}{x \times x^2} \times x^2$$

$$= 0$$

differentiable at 0

$$f'(x) = \begin{cases} \frac{3x^2}{1+x^3} \sin \left(\frac{1}{x} \right) - \frac{x}{x^2} \frac{\ln(1+x^3)}{x} \cos \left(\frac{1}{x} \right) & ; x > 0 \\ 0 & ; x \leq 0 \end{cases}$$

$$f'(0^-) = 0 = f'(0) = f'(0^+) \Rightarrow \text{Continuous at } x = 0$$

$$(B) \quad g(x) = \begin{cases} \ln^2(1+x) \cdot \sin\left(\frac{1}{x}\right) & ; \quad x < 0 \\ 0 & ; \quad x \leq 0 \end{cases}$$

$$g(0^+) = 0 = g(0) = g(0^-)$$

$\Rightarrow$ Continuous at 0;

$$\text{LHD} = 0$$

$$\text{RHD} = \lim_{x \rightarrow 0} \frac{\ln^2(1+x) \sin \frac{1}{x} - 0}{x \times \frac{n}{x}}$$

$$= 0$$

Differentiable at $x = 0$

$$g'(x) = \begin{cases} \frac{2 \ln(1+x)}{1+x} \sin \frac{1}{x} & ; \quad x > 0 \\ 0 & ; \quad x \leq 0 \end{cases}$$

Derivative is not continuous

$$(C) \quad u(x) = \begin{cases} \ln\left(1 + \frac{\sin u}{2}\right) & ; \quad x > 0 \\ 0 & ; \quad x \leq 0 \end{cases}$$

$$u(0^+) = 0 = u(0) = u(0^-)$$

$\Rightarrow$ continuous

$$\text{LHD} = 0$$

$$\text{RHD} = \lim_{x \rightarrow 0} \left[\frac{\ln\left(1 + \frac{\sin x}{2}\right) - 0}{x \times \frac{\sin x}{2}} \right] \left(\frac{\sin x}{2} \right)$$

$$= \frac{1}{2}$$

not differentiable at $x = 0$

$$(D) v(x) = \lim_{t \rightarrow 0} \frac{2x}{11} \tan^{-1} \left(\frac{2}{t^2} \right)$$

$$= \frac{2x}{\pi} \lim_{t \rightarrow 0} \tan^{-1} \left(\frac{2}{t^2} \right)$$

$$= \frac{2x}{\pi} \times \frac{\pi}{2}$$

$$= x$$

48. Column-I contains function defined on R and Column-II contains their properties. Match them.

Column-I

Column-II

$$(A) \lim_{n \rightarrow \infty} \left(\frac{1 + \tan \frac{\pi}{2n}}{1 + \sin \frac{\pi}{3n}} \right)^n \text{ equal}$$

$$(P) e$$

$$(B) \lim_{x \rightarrow 0^+} \frac{1}{(1 + \operatorname{cosec} x)^{\frac{1}{\ln(\sin x)}}} \text{ equals}$$

$$(Q) e^2$$

$$(C) \lim_{x \rightarrow 0} \left(\frac{2}{\pi} \cos^{-1} x \right)^{1/x} \text{ equals}$$

$$(R) e^{-2/\pi}$$

$$(S) e^{\pi/6}$$

कॉलम-I में R में परिभाषित फलन दिये गए हैं तथा कॉलम-II में उनके गुणधर्म। सुमेलित कीजिये।

Column-I

Column-II

$$(A) \lim_{n \rightarrow \infty} \left(\frac{1 + \tan \frac{\pi}{2n}}{1 + \sin \frac{\pi}{3n}} \right)^n \text{ बराबर है}$$

$$(P) e$$

$$(B) \lim_{x \rightarrow 0^+} \frac{1}{(1 + \operatorname{cosec} x)^{\frac{1}{\ln(\sin x)}}} \text{ बराबर है}$$

$$(Q) e^2$$

$$(C) \lim_{x \rightarrow 0} \left(\frac{2}{\pi} \cos^{-1} x \right)^{1/x} \text{ बराबर है}$$

$$(R) e^{-2/\pi}$$

$$(S) e^{\pi/6}$$

Ans. (A) S; (B) P; (C) R

Sol. (A) $\lim_{n \rightarrow \infty} \left(\frac{1 + \tan \frac{\pi}{2n}}{1 + \sin \frac{\pi}{3n}} \right)^{\frac{\pi}{6n}} = \theta$

$$= 1^\infty$$

$$\Rightarrow (e)^{\lim_{n \rightarrow \infty} n \left(\frac{1 + \tan 3\theta}{1 + \sin 2\theta} - 1 \right)}$$

$$\Rightarrow (e)^{\lim_{n \rightarrow \infty} n \left(\frac{1 + \frac{3t - t^3}{1 - 3t^2} - \frac{2^t}{1 + t^2} - 1}{1 + \sin 2\theta} \right)} \quad (t = \tan x)$$

$$\Rightarrow (e)^{\lim_{n \rightarrow \infty} n \cdot \frac{\tan \frac{\pi}{6n}}{\frac{\pi}{6n}} \left[\frac{3 - t^2}{1 - 3t^2} - \frac{2}{1 + t^2} \right] \cdot \frac{1}{\left(1 + \sin \frac{\pi}{3n} \right)} \times \frac{\pi}{6n}}$$

$$\Rightarrow (e)^{\frac{\pi}{6} (1)(3-2)} = (e)^{\frac{\pi}{6}}$$

(B) $\lim_{x \rightarrow 0^+} \frac{1}{(1 + \operatorname{cosec} x)^{\frac{1}{\ln(\sin x)}}}$

$$\lim_{x \rightarrow 0^+} (e)^{\frac{\frac{1}{\ln(1 + \operatorname{cosec} x)}}{\frac{1}{\ln(\sin x)}}} = \lim_{x \rightarrow 0^+} e^{\frac{\ln \sin x}{\ln(1 + \operatorname{cosec} x)}}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} (e)^{\frac{\frac{\cos x}{\sin x}}{\frac{\cos \operatorname{cosec} \cot x}{1 + \cos \operatorname{cosec}}}}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} (e)^{\frac{\cos x}{\sin x} \times \frac{1 + \cos \operatorname{cosec} x}{-\cos \operatorname{cosec} \cot x}}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} (e)^{1 + \sin x}$$

$$= e^1$$

(C) $\lim_{x \rightarrow 0^+} \left[\frac{2}{\pi} \cos^{-1} x \right]^{\frac{1}{x}}$

$$\Rightarrow \lim_{x \rightarrow 0^+} \left[\frac{2}{\pi} \left(\frac{\pi}{2} - \sin^{-1} x \right) \right]^{\frac{1}{x}}$$

$$\Rightarrow (e)^{\lim_{x \rightarrow 0^+} \frac{1}{x} \left[1 - \frac{2}{\pi} \sin^{-1} x - 1 \right]}$$

$$= (e)^{-\frac{2}{\pi}}$$

Exercise J - A

1. Let $f(x) = x \sin \pi x$, $x > 0$. Then for all natural numbers n , $f(x)$ vanishes at : [JEE(Adv.)-2013]

- (A) a unique point in the interval $\left(n, n + \frac{1}{2}\right)$ (B) a unique point in the interval $\left(n + \frac{1}{2}, n + 1\right)$
 (C) a unique point in the interval $(n, n + 1)$ (D) two points in the interval $(n, n + 1)$

माना कि $f(x) = x \sin \pi x$, $x > 0$ तब सभी धन-पूर्णाकों n के लिए $f(x)$ निम्न पर शून्य होता है : [JEE(Adv.)-2013]

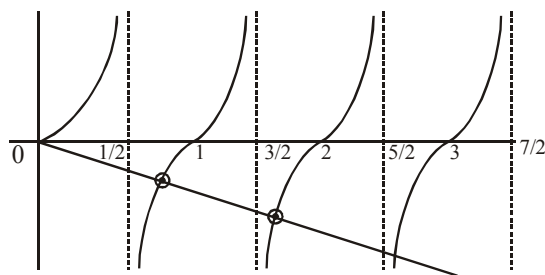
- (A) अंतराल $\left(n, n + \frac{1}{2}\right)$ में एकमात्र एक बिन्दु पर (B) अंतराल $\left(n + \frac{1}{2}, n + 1\right)$ में एकमात्र एक बिन्दु पर
 (C) अंतराल $(n, n + 1)$ में एकमात्र एक बिन्दु पर (D) अंतराल $(n, n + 1)$ में दो बिन्दुओं पर

Ans. BC

Sol. $f'(x) = \sin \pi x + \pi x \cos \pi x = 0$

$$\tan \pi x = -\pi x$$

$$y = \tan x \pi \text{ \& } y = -\pi x$$



intersection point lies in

$$\left(\frac{1}{2}, 1\right) \cup \left(\frac{3}{2}, 2\right) \cup \left(\frac{5}{2}, 3\right) \dots$$

option (B) is correct for $\left(n + \frac{1}{2}, n + 1\right)$

as well $(n, (n + 1))$ because root lies in

$$(0, 1) \cup (1, 2) \cup (2, 3)$$

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then : [JEE(Adv.)-2016]

माना कि $f : \mathbb{R} \rightarrow \mathbb{R}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ और $h : \mathbb{R} \rightarrow \mathbb{R}$ ऐसे अवकलनीय फलन हैं कि सभी $x \in \mathbb{R}$ के लिए $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ और $h(g(g(x))) = x$ हैं। तब : [JEE(Adv.)-2016]

- (A) $g'(2) = \frac{1}{15}$ (B) $h'(1) = 666$ (C) $h(0) = 16$ (D) $h(g(3)) = 36$

Ans. BC

Sol. Here, $g(f(x)) = x \Rightarrow g(x) = f^{-1}(x)$

$$h(g(g(x))) = x \Rightarrow h(g(x)) = f(x) \quad (i)$$

$$h(g(g(x))) = x \Rightarrow h(x) = f(f(x)) \quad (ii)$$

$$(A) f(g(x)) = x \Rightarrow f'(g(x)) g'(x) = 1 \therefore g'(2) = \frac{1}{f'(g(2))} = \frac{1}{f'(0)} = \frac{1}{3}$$

$$(B) h'(x) = f'(f(x)) \cdot f'(x) \quad (\because \text{using (ii)})$$

$$\text{Put } x = 1 \Rightarrow h'(1) = 666$$

$$(C) h(0) = f(f(0)) = 16$$

$$(D) h(g(3)) = f(3) \quad (\because \text{using (i)})$$

Paragraph for Question Nos. 3 to 4

Answer the following by appropriately matching the lists based on the information given in the paragraph. Let $f(x) = \sin(\pi \cos x)$ and $g(x) = \cos(2\pi \sin x)$ be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order :

$$X = \{x : f(x) = 0\}, \quad Y = \{x : f'(x) = 0\}$$

$$Z = \{x : g(x) = 0\}, \quad W = \{x : g'(x) = 0\}$$

Column-I contains the sets X, Y, Z and W. Column-II contains some information regarding these sets.

Column-I

(I) X

(II) Y

(III) Z

(IV) W

Column-II

$$(P) \supseteq \left\{ \frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi \right\}$$

(Q) an arithmetic progression

(R) NOT an arithmetic progression

$$(S) \supseteq \left\{ \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6} \right\}$$

$$(T) \supseteq \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \pi \right\}$$

$$(U) \supseteq \left\{ \frac{\pi}{6}, \frac{3\pi}{4} \right\}$$

(Que. 3 से 4) अनुच्छेद में दी गई जानकारी के आधार पर सूचियों का उचित मिलान करके प्रश्न का उत्तर दें। माना कि $f(x) = \sin(\pi \cos x)$ और $g(x) = \cos(2\pi \sin x)$ दो फलन (function) है जो $x > 0$ में परिभाषित हैं।

$$X = \{x : f(x) = 0\}, \quad Y = \{x : f'(x) = 0\}$$

$$Z = \{x : g(x) = 0\}, \quad W = \{x : g'(x) = 0\}$$

सूची-I में X, Y, Z और W समुच्चय है। सूची-II में इन समुच्चयों के बारे में कुछ सूचनाएं हैं।

Column-I

Column-II

- (I) X (P) $\supseteq \left\{ \frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi \right\}$
 (II) Y (Q) समान्तर श्रेणी (an arithmetic progression)
 (III) Z (R) समान्तर श्रेणी नहीं है (NOT an arithmetic progression)
 (IV) W (S) $\supseteq \left\{ \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6} \right\}$
 (T) $\supseteq \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \pi \right\}$
 (U) $\supseteq \left\{ \frac{\pi}{6}, \frac{3\pi}{4} \right\}$

3. Which of the following is the only CORRECT combination : [JEE(Adv.)-2019]

निम्न में से कौन सा एकमात्र संयोजन सही है : [JEE(Adv.)-2019]

- (A) (I), (Q), (U) (B) (II), (R), (S) (C) (I), (P), (R) (D) (II), (Q), (T)

Ans. D

4. Which of the following is the only CORRECT combination : [JEE(Adv.)-2019]

निम्न में से कौन सा एकमात्र संयोजन सही है : [JEE(Adv.)-2019]

- (A) (III), (P), (Q), (U) (B) (III), (R), (U) (C) (IV), (P), (R), (S) (D) (IV), (Q), (T)

Ans. C

Sol. X: $f(x) = 0$

$$\begin{aligned}\sin(\pi \cos x) &= 0 \\ \pi \cos x &= n\pi \quad n \in I \\ \cos x &= n \quad n \in I \\ \therefore \cos x &= 0 \text{ or } \cos x = -1 \quad \text{or } \cos x = 1\end{aligned}$$

$$x = (2n+1)\frac{\pi}{2} \quad x = n\pi$$

$$\therefore x \in \left\{ n\pi, \left(n + \frac{1}{2}\right)\pi \right\}$$

$$\therefore x = \left\{ \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \dots \right\}$$

Y:

$$\begin{aligned}f(x) &= 0 \\ \cos(\pi \cos x) \cdot \pi(-\sin x) &= 0 \\ \cos(\pi \cos x) &= 0 \quad \text{or } \sin x = 0\end{aligned}$$

$$\pi \cos x = (2n+1) \frac{\pi}{2} \quad x = n\pi$$

$$\cos x = \frac{(2n+1)}{2}$$

$$\therefore \cos x = \frac{1}{2}, \frac{-1}{2}$$

$$x = 2n\pi \pm \frac{\pi}{3} = \left(2n + \frac{1}{3}\right)\pi, \left(2n - \frac{1}{3}\right)\pi$$

$$x = 2n\pi \pm \frac{2\pi}{3} = \left(2n + \frac{2}{3}\right)\pi, \left(2n - \frac{2}{3}\right)\pi$$

$$y = \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \dots \right\}$$

Z: $g(x) = 0$
 $\cos(2\pi \sin x) = 0$
 $2\pi \sin x = (2n+1)\pi/2 \quad n \in I$

$$\sin x = \frac{(2n+1)}{4}$$

$$\sin x = \frac{1}{4}, \frac{-1}{4}, \frac{3}{4}, \frac{-3}{4}$$

$$x = n\pi \pm (-1)^n \sin^{-1}(1/4)$$

$$\text{or } x = n\pi \pm (-1)^n \sin^{-1}(3/4)$$

W: $g'(x) = 0$
 $-\sin(2\pi \sin x) \cdot 2\pi \cos x = 0$
 $\sin(2\pi \sin x) = 0 \text{ or } \cos x = 0$

$$2\pi \sin x = n\pi \quad x = (2n+1) \frac{\pi}{2}$$

$$\sin x = n/2$$

$$\therefore \sin x = 0, \frac{1}{2}, \frac{-1}{2}, 1, -1$$

$$\therefore x = n\pi$$

$$\text{or } x = n\pi \pm (-1)^n \frac{\pi}{6}$$

$$\text{or } x = 2n\pi \pm \frac{\pi}{2}$$

5. Let b be a nonzero real number. Suppose $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f(0) = 1$. If the derivative f' of f satisfies the equation $f'(x) = \frac{f(x)}{b^2 + x^2}$.

for all $x \in \mathbb{R}$, then which of the following statements is/are TRUE?

[JEE(Adv.)-2020]

- (A) If $b > 0$, then f is an increasing function (B) If $b < 0$, then f is a decreasing function
(C) $f(x)f(-x) = 1$ for all $x \in \mathbb{R}$ (D) $f(x) - f(-x) = 0$ for all $x \in \mathbb{R}$

माना कि b एक शून्येतर वास्तविक संख्या है। मान लीजिये कि $f: \mathbb{R} \rightarrow \mathbb{R}$ एक ऐसा अवकलनीय फलन है जिसके लिये $f(0) = 1$ है।

यदि f का अवकलज f' समीकरण $f'(x) = \frac{f(x)}{b^2 + x^2}$ को सभी $x \in \mathbb{R}$ के लिये संतुष्ट करता है, तब निम्न में से कौन सा (से) कथन सही

ग(है)?

[JEE(Adv.)-2020]

- (A) यदि $b > 0$ है, तब f एक वर्धमान फलन है (B) यदि $b < 0$ है, तब f एक ह्रासमान फलन है
(C) सभी $x \in \mathbb{R}$ के लिए $f(x)f(-x) = 1$ है (D) सभी $x \in \mathbb{R}$ के लिए $f(x) - f(-x) = 0$ है

Ans. AC

Sol. f is non zero

$$\int \frac{f'(x)}{f(x)} dx = \int \frac{1}{x^2 + b^2} dx$$

$$\ln(f(x)) = \frac{1}{b} \tan^{-1} \frac{x}{b} + c$$

$$0 = 0 + c \quad \{\because f(0) = 1\}$$

$$f(x) = e^{\frac{1}{b} \tan^{-1} \frac{x}{b}}$$

$$f(-x) = e^{\frac{-1}{b} \tan^{-1} \frac{x}{b}}$$

(A) $f(x)$ is positive function for all $x \in \mathbb{R}$

$$f'(x) = \frac{f(x)}{x^2 + b^2}$$

$f'(x)$ is also positive function

so $f(x)$ is increasing function

$$(d) f(x) - f(-x) = e^{\frac{1}{b} \tan^{-1} \frac{x}{b}} - e^{\frac{-1}{b} \tan^{-1} \frac{x}{b}} \neq 0$$

for all $x \in \mathbb{R}$

$$(c) f(x)f(-x) = 1$$

$$e^0 = 1$$

$$1 = 1$$

6. Let α be a positive real number. Let $f : R \rightarrow R$ and $g : (\alpha, \infty) \rightarrow R$ be the functions defined by $f(x) = \sin\left(\frac{\pi x}{12}\right)$

and $g(x) = \frac{2 \log_e (\sqrt{x} - \sqrt{\alpha})}{\log_e (e^{\sqrt{x}} - e^{\sqrt{\alpha}})}$. Then the value of $\lim_{x \rightarrow \alpha^+} f(g(x))$ is _____. [JEE(Adv.)-2022]

माना कि α एक धनात्मक वास्तविक संख्या (positive real number) है। माना कि फलनों (functions) $f : R \rightarrow R$ एवं

$g : (\alpha, \infty) \rightarrow R$ को $f(x) = \sin\left(\frac{\pi x}{12}\right)$ एवं $g(x) = \frac{2 \log_e (\sqrt{x} - \sqrt{\alpha})}{\log_e (e^{\sqrt{x}} - e^{\sqrt{\alpha}})}$ द्वारा परिभाषित किया जाता है। तब $\lim_{x \rightarrow \alpha^+} f(g(x))$

का मान _____ है।

[JEE(Adv.)-2022]

Ans. 00.50

Sol. $\lim_{x \rightarrow \alpha^+} g(x) = \lim_{x \rightarrow \alpha^+} \frac{\frac{2}{\sqrt{x} - \sqrt{\alpha}} \left(\frac{1}{2\sqrt{x}} \right)}{\frac{1}{e^{\sqrt{x}} - e^{\sqrt{\alpha}}} \left(\frac{1}{2\sqrt{x}} \cdot e^{\sqrt{x}} \right)}$ (Using L'Hospital Rule)

$$= \lim_{x \rightarrow \alpha^+} \frac{e^{\sqrt{x}} - e^{\sqrt{\alpha}}}{\sqrt{x} - \sqrt{\alpha}} \cdot \frac{1}{e^{\sqrt{x}}} \cdot 2$$

$$= \lim_{x \rightarrow \alpha^+} \frac{e^{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}}}{\frac{1}{2\sqrt{x}}} \cdot \frac{2}{e^{\sqrt{x}}} = 2$$

$$\lim_{x \rightarrow \alpha^+} f(g(x)) = f\left(\lim_{x \rightarrow \alpha^+} g(x)\right) = \sin \frac{\pi}{6} = \frac{1}{2} = 00.50$$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. If $y = (x + \sqrt{1+x^2})^n$ then $(1+x^2)y_2 + xy_1 =$
(A) ny^2 (B) n^2y (C) n^2y^2 (D) None of these
- यदि $y = (x + \sqrt{1+x^2})^n$ है, तो $(1+x^2)y_2 + xy_1 =$
(A) ny^2 (B) n^2y (C) n^2y^2 (D) इनमें से कोई नहीं

Ans. B

Sol. $y = (x + \sqrt{1+x^2})^n$

$$y_1 = n(x + \sqrt{1+x^2})^{n-1} \left(1 + \frac{2x}{2\sqrt{1+x^2}}\right)$$

$$y_1 = \frac{n(x + \sqrt{1+x^2})^n}{\sqrt{1+x^2}}$$

$$y_2 = \frac{n^2(x + \sqrt{1+x^2})^{n-1} \left(1 + \frac{x}{\sqrt{1+x^2}}\right) \sqrt{1+x^2} - n(x + \sqrt{1+x^2})^n \frac{2x}{2\sqrt{1+x^2}}}{(1+x^2)}$$

$$(1+x^2)y_2 = n^2(x + \sqrt{1+x^2})^n - \frac{n(x + \sqrt{1+x^2})^n}{\sqrt{1+x^2}} \cdot x$$

$$(1+x^2)y_2 = n^2y - y_1x$$

$$(1+x^2)y_2 + xy_1 = n^2y$$

2. Let $u(x)$ and $v(x)$ are differentiable functions such that $\frac{u(x)}{v(x)} = 7$. If $\frac{u'(x)}{v'(x)} = p$ and $\left(\frac{u(x)}{v(x)}\right)' = q$, then $\frac{p+q}{p-q}$ has the value equal to :

यदि $u(x)$ और $v(x)$ अवकलनीय फलन इस प्रकार है कि $\frac{u(x)}{v(x)} = 7$ यदि $\frac{u'(x)}{v'(x)} = p$ और $\left(\frac{u(x)}{v(x)}\right)' = q$, तब $\frac{p+q}{p-q}$ का मान

ckkj :

(A) 1

(B) 0

(C) 7

(D) -7

Ans. A

Sol. $u(x) = 7v(x) \Rightarrow u'(x) = 7v'(x) \Rightarrow p = 7$ (given)

again $\frac{u(x)}{v(x)} = 7 \Rightarrow \left(\frac{u(x)}{v(x)}\right)' = 0 \Rightarrow q = 0$; now $\frac{p+q}{p-q} = \frac{7+0}{7-0} = 1$ **Ans.**

3. If $x = t^3 + t + 5$ and $y = \sin t$ then $\frac{d^2y}{dx^2} =$

यदि $x = t^3 + t + 5$ तथा $y = \sin t$ तब $\frac{d^2y}{dx^2} =$

(A) $-\frac{(3t^2 + 1) \sin t + 6t \cos t}{(3t^2 + 1)^3}$

(B) $\frac{(3t^2 + 1) \sin t + 6t \cos t}{(3t^2 + 1)^2}$

(C) $-\frac{(3t^2 + 1) \sin t + 6t \cos t}{(3t^2 + 1)^2}$

(D) $\frac{\cos t}{3t^2 + 1}$

Ans. A

Sol. $x = t^3 + t + 5$

$y = \sin t$

$$\frac{dx}{dt} = 3t^2 + 1$$

$$\frac{dy}{dt} = \cos t$$

$$\frac{dy}{dx} = \frac{\cos t}{3t^2 + 1}$$

$$\frac{d^2y}{dx^2} = \frac{-\sin t \frac{dt}{dx} (3t^2 + 1) - \cos t \left(6t \frac{dt}{dx}\right)}{(3t^2 + 1)^2}$$

$$= \frac{-\frac{\sin t}{3t^2+1}(3t^2+1) - \frac{6t \cos t}{1+3t^2}}{(3t^2+1)^2}$$

$$= -\frac{\sin t(3t^2+1) + 6t \cos t}{(3t^2+1)^3}$$

4. If $f(x) = \frac{a + \sqrt{a^2 - x^2} + x}{\sqrt{a^2 - x^2} + a - x}$ where $a > 0$ and $x < a$, then $f'(0)$ has the value equal to :

यदि $f(x) = \frac{a + \sqrt{a^2 - x^2} + x}{\sqrt{a^2 - x^2} + a - x}$ जहाँ $a > 0$ और $x < a$, तब $f'(0)$ का मान होगा :

- (A) $\sqrt{a}$ (B) a (C) $\frac{1}{\sqrt{a}}$ (D) $\frac{1}{a}$

Ans. D

Sol. $y = \frac{(a+x) + \sqrt{a-x} \cdot \sqrt{a+x}}{(a-x) + \sqrt{a-x} \cdot \sqrt{a+x}}$

$$y = \frac{\sqrt{a+x}(\sqrt{a+x} + \sqrt{a-x})}{\sqrt{a-x}(\sqrt{a+x} + \sqrt{a-x})} = \left(\frac{a+x}{a-x}\right)^{1/2}$$

$$\frac{dy}{dx} = \frac{1}{2} \sqrt{\frac{a-x}{a+x}} \left(\frac{(a-x) + (a+x)}{(a-x)^2} \right) = \frac{1}{2} \frac{\sqrt{a-x}}{\sqrt{a+x}} \times \frac{2a}{(a-x)^2}; \therefore \left. \frac{dy}{dx} \right|_{x=0} = \frac{1}{a} \text{ Ans.}$$

5. Suppose that $f(0) = 0$ and $f'(0) = 2$, and let $g(x) = f(-x + f(f(x)))$. The value of $g'(0)$ is equal to :

माना कि $f(0) = 0$ और $f'(0) = 2$, तथा $g(x) = f(-x + f(f(x)))$. $g'(0)$ का मान बराबर :

- (A) 0 (B) 1 (C) 6 (D) 8

Ans. C

Sol. $g(x) = f(-x + f(f(x)))$; $f(0) = 0$; $f'(0) = 2$

$$g'(x) = f'(-x + f(f(x))) \cdot [-1 + f'(f(x)) \cdot f'(x)]$$

$$g'(0) = f'(f(0)) \cdot [-1 + f'(0) \cdot f'(0)]$$

$$= f'(0) [-1 + (2)(2)]$$

$$= (2)(3) = 6 \text{ Ans.}$$

6. If $\ln(3 \sin x - 4 \cos x + 7 + 5y) = (\sin^2 x)y$, then $y'(\pi)$ is equal to :

यदि $\ln(3 \sin x - 4 \cos x + 7 + 5y) = (\sin^2 x)y$ हो, तब $y'(\pi)$ बराबर होगा :

- (A) $\frac{5}{3}$ (B) 0 (C) $\frac{3}{5}$ (D) $\frac{-3}{5}$

Ans. C

Sol. We have $\ln(3 \sin x - 4 \cos x + 7 + 5y) = (\sin^2 x)y$ (1)

Put $x = \pi$ in equation (1), we get $\ln(11 + 5y) = 0 \Rightarrow 11 + 5y = 1 \Rightarrow 5y = -10$

$\therefore y = -2$. So, $(\pi, -2)$ lies on the given curve.

Now on differentiating both the sides of equation (1) w.r.t. x , we get

$$\frac{1}{(3 \sin x - 4 \cos x + 7 + 5y)} \times \left(3 \cos x + 4 \sin x + 5 \frac{dy}{dx} \right) = (\sin^2 x) \frac{dy}{dx} + (\sin 2x)y$$

$$\text{As } (\pi, -2) \text{ satisfy it, we get } 1 \times \left(-3 + 0 + 5 \frac{dy}{dx} \right) = 0$$

$$\therefore \left. \frac{dy}{dx} \right|_{(\pi, -2)} = \frac{3}{5} \Rightarrow y'(\pi) = \frac{3}{5}. \text{ Ans.}$$

7. If $f'(x) = -f(x)$ and $g(x) = f'(x)$ and $F(x) = \left(f\left(\frac{x}{2}\right) \right)^2 + \left(g\left(\frac{x}{2}\right) \right)^2$ and given that $F(5) = 5$, then $F(10)$ is equal to :

[JEE(Adv.)-2006]

यदि $f'(x) = -f(x)$ एवं $g(x) = f'(x)$ तथा $F(x) = \left(f\left(\frac{x}{2}\right) \right)^2 + \left(g\left(\frac{x}{2}\right) \right)^2$ एवं $F(5) = 5$ हो तब $F(10)$ का मान है :

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- (A) 5 (B) 10 (C) 0 (D) 15

Ans. A

Sol. $\therefore g(x) = f'(x)$ (i)

$\Rightarrow g'(x) = f''(x) = -f(x)$ (ii)

$$F(x) = \left(f\left(\frac{x}{2}\right) \right)^2 + \left(g\left(\frac{x}{2}\right) \right)^2$$

$$\Rightarrow F'(x) = 2f(x/2) \cdot f'(x/2) \cdot \frac{1}{2} + 2g(x/2) \cdot g'(x/2) \cdot \frac{1}{2}$$

$$\Rightarrow F'(x) = f\left(\frac{x}{2}\right) f'\left(\frac{x}{2}\right) + g\left(\frac{x}{2}\right) g'\left(\frac{x}{2}\right)$$

$$\Rightarrow F'(x) = f\left(\frac{x}{2}\right) \cdot f'\left(\frac{x}{2}\right) - f'\left(\frac{x}{2}\right) f\left(\frac{x}{2}\right) \quad \{\text{Using (i) \& (ii)}\}$$

$$\Rightarrow F'(x) = 0$$

$$\Rightarrow F(x) = \text{constant} \Rightarrow F(5) = F(10) = 5$$

8. If $y = (\tan^{-1}x)^2$ then $(x^2 + 1)^2 D^2y + 2x(x^2 + 1) Dy$ has the value equal to :

- (A) 0 (B) 2 (C) 4 (D) None of these

यदि $y = (\tan^{-1}x)^2$ तब $(x^2 + 1)^2 D^2y + 2x(x^2 + 1) Dy$ बराबर :

- (A) 0 (B) 2 (C) 4 (D) इनमें से कोई नहीं

Ans. B

Sol. $(1 + x^2) y_1 = 2 \tan^{-1}x$

$$(1 + x^2)^2 \cdot y_1^2 = 4(\tan^{-1}x)^2 = 4y$$

$$(1 + x^2)^2 \cdot 2y_1 y_2 + y_1^2 \cdot 2(1 + x^2) \cdot 2x = 4y_1$$

$$(1 + x^2)^2 y_2 + 2x(1 + x^2)y_1 = 2 \quad \text{Ans.]}$$

Alternatively: $y = (\tan^{-1}x)^2$

$$y_1 = \frac{2 \tan^{-1}x}{1+x^2}; \quad (1+x^2)y_1 = 2 \tan^{-1}x; \quad (1+x^2)y_2 + 2xy_1 = \frac{2}{1+x^2}$$

$$(1+x^2)^2 y_2 + 2x(1+x^2)y_1 = 2$$

Hence $\boxed{k=2}$ Ans.

9. Let $g(x)$ be an antiderivative for $f(x)$. Then $\ln(1+(g(x))^2)$ is an antiderivative for :

- (A) $\frac{2f(x)g(x)}{1+(f(x))^2}$ (B) $\frac{2f(x)g(x)}{1+(g(x))^2}$ (C) $\frac{2f(x)}{1+(f(x))^2}$ (D) None of these

यदि $g(x)$, $f(x)$ का समाकलन है तब $\ln(1+(g(x))^2)$ समाकलन निम्न में से किसका समाकलन होगा :

- (A) $\frac{2f(x)g(x)}{1+(f(x))^2}$ (B) $\frac{2f(x)g(x)}{1+(g(x))^2}$ (C) $\frac{2f(x)}{1+(f(x))^2}$ (D) इनमें से कोई नहीं

Ans. B

Sol. Given $\int f(x) dx = g(x) \Rightarrow g'(x) = f(x)$

$$\text{now } \frac{d}{dx} \left(\ln(1+g^2(x)) \right) = \frac{2g(x)g'(x)}{1+g^2(x)} = \frac{2f(x)g(x)}{1+g^2(x)}$$

10. If $y = (A + Bx)e^{mx} + (m-1)^{-2}e^x$ then $\frac{d^2y}{dx^2} - 2m\frac{dy}{dx} + m^2y$ is equal to :

यदि $y = (A + Bx)e^{mx} + (m-1)^{-2}e^x$ है, तो $\frac{d^2y}{dx^2} - 2m\frac{dy}{dx} + m^2y$ बराबर होगा :

- (A) e^x (B) e^{mx} (C) e^{-mx} (D) $e^{(1-m)x}$

Ans. A

Sol. $y = (A + Bx)e^{mx} + (m-1)^{-2} \cdot e^x$
 $y \cdot e^{-mx} = (A + Bx) + (m-1)^{-2} \cdot e^{(1-m)x}$
 $e^{-mx} \cdot y_1 - m y e^{-mx} = B - (m-1)^{-1} \cdot e^{-(m-1)x}$
 $e^{-mx} \cdot y_2 - y_1 e^{-mx} \cdot m - m[e^{-mx} \cdot y_1 - y e^{-mx} \cdot m] = e^{-(m-1)x}$
 $e^{-mx} \cdot y_1 - m_2 y_1 e^{-mx} + m y \cdot e^{-mx} = e^{-(m-1)x}$
 $y_2 - 2m y_1 + m y = e^x$

11. Let $f(x) = x^n$, n being a non-negative integer. The number of values of n for which $f'(p+q) = f'(p) + f'(q)$ is valid for all $p, q > 0$ is :

- (A) 0 (B) 1 (C) 2 (D) None of these

माना $f(x) = x^n$, n अऋणात्मक पूर्णांक है तब n के मानों की संख्या जिसके लिए $f'(p+q) = f'(p) + f'(q)$ सभी $p, q > 0$ के लिए $1 \in \mathbb{R}$ $g \in \mathbb{S}$

- (A) 0 (B) 1 (C) 2 (D) इनमें से कोई नहीं

Ans. C

Sol.

12. The ends A and B of a rod of length $\sqrt{5}$ are sliding along the curve $y = 2x^2$. Let x_A and x_B be the x-coordinate of the ends. At the moment when A is at (0, 0) and B is at (1, 2) the derivative $\frac{dx_B}{dx_A}$ has the value equal to :

एक $\sqrt{5}$ लम्बाई की छड़ के अन्तिम सिरे A व B वक्र $y = 2x^2$ के अनुदिश फिसल रहे हैं यदि x_A व x_B सिरे के x-निर्देशांक हो तो $\frac{dx_B}{dx_A}$ का मान क्या होगा जबकि A (0, 0) व B (1, 2) पर है :

- (A) $1/3$ (B) $1/5$ (C) $1/8$ (D) $1/9$

Ans. D

Sol. We have $y = 2x^2$

$$(AB)^2 = (x_B - x_A)^2 + (2x_B^2 - 2x_A^2)^2 = 5$$

$$\text{or } (x_B - x_A)^2 + 4(x_B^2 - x_A^2)^2 = 5$$

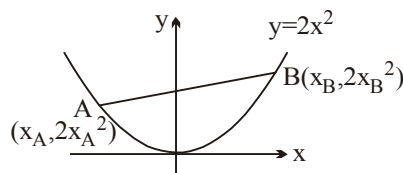
differentiating w.r.t. x_A and denoting $\frac{dx_B}{dx_A} = D$

$$2(x_B - x_A)(D - 1) + 8(x_B^2 - x_A^2)(2x_B D - 2x_A) = 0$$

$$\text{put } x_A = 0; x_B = 1$$

$$2(1 - 0)(D - 1) + 8(1 - 0)(2D - 0) = 0$$

$$2D - 2 + 16D = 0 \Rightarrow D = 1/9 \text{ Ans.}$$



13. The derivative of the function, $f(x) = \cos^{-1} \left\{ \frac{1}{\sqrt{13}} (2 \cos x - 3 \sin x) \right\} + \sin^{-1} \left\{ \frac{1}{\sqrt{13}} (2 \cos x + 3 \sin x) \right\}$ w.r.t.

$\sqrt{1 + x^2}$ at $x = \frac{3}{4}$ is :

दिये गये फलन $f(x) = \cos^{-1} \left\{ \frac{1}{\sqrt{13}} (2 \cos x - 3 \sin x) \right\} + \sin^{-1} \left\{ \frac{1}{\sqrt{13}} (2 \cos x + 3 \sin x) \right\}$ का अवकलन $\sqrt{1 + x^2}$ के

1 कि $x = \frac{3}{4}$ पर होगा :

- (A) $\frac{3}{2}$ (B) $\frac{5}{2}$ (C) $\frac{10}{3}$ (D) 0

Ans. C

Sol. Put $\cos \phi = \frac{2}{\sqrt{13}}$; $\sin \phi = \frac{3}{\sqrt{13}}$; $\tan \phi = \frac{3}{2}$

$$y = \cos^{-1} \{ \cos(x + \phi) \} + \sin^{-1} \{ \cos(x - \phi) \}$$

$$= \cos^{-1} \{ \cos(x + \phi) + \frac{\pi}{2} - \cos^{-1} \{ \cos(\phi - x) \} \} \text{ (think !)}$$

$$= x + \phi + \frac{\pi}{2} - \phi + x$$

$$y = 2x + \frac{\pi}{2} ; z = \sqrt{1+x^2}$$

now compute $\frac{dy}{dz}$

14. Let $f(x) = \sin^3 x + \sin^3 \left(x + \frac{2\pi}{3} \right) + \sin^3 \left(x + \frac{4\pi}{3} \right)$ then the primitive of $f(x)$ w.r.t. x is (where C is an arbitrary constant) :

यदि $f(x) = \sin^3 x + \sin^3 \left(x + \frac{2\pi}{3} \right) + \sin^3 \left(x + \frac{4\pi}{3} \right)$ तब $f(x)$ का x के सापेक्ष समाकल है (जहाँ C स्वेच्छ अचर है) :

(A) $-\frac{3 \sin 3x}{4} + C$ (B) $-\frac{3 \cos 3x}{4} + C$ (C) $\frac{\sin 3x}{4} + C$ (D) $\frac{\cos 3x}{4} + C$

Ans. D

Sol. Note that $\sin x + \sin \left(x + \frac{2\pi}{3} \right) + \sin \left(x + \frac{4\pi}{3} \right) = 0$

$$\Rightarrow \sin^3 x + \sin^3 \left(x + \frac{2\pi}{3} \right) + \sin^3 \left(x + \frac{4\pi}{3} \right) = -\frac{3}{4} \sin 3x \quad (a+b+c=0 \Rightarrow a^3+b^3+c^3=3abc)$$

$$\therefore -\frac{3}{4} \int \sin 3x \, dx = \frac{\cos 3x}{4} + C \text{ Ans.}$$

15. People living at Mars, instead of the usual definition of derivative $Df(x)$, define a new kind of derivative, $Df(x)$ by the formula $Df(x) = \lim_{h \rightarrow 0} \frac{f^2(x+h) - f^2(x)}{h}$ where $f^2(x)$ means $[f(x)]^2$. If $f(x) = x \ln x$ then $D * f(x) \big|_{x=e}$ has the value :

यदि अवकलन की सामान्य परिभाषा $Df(x)$ के स्थान पर अवकलन की नई परिभाषा $Df(x)$ सूत्र $Df(x) = \lim_{h \rightarrow 0} \frac{f^2(x+h) - f^2(x)}{h}$

से दी जाती है। जहाँ $f^2(x)$ का अर्थ $[f(x)]^2$ है। यदि $f(x) = x \ln x$ तब $D^*f(x)|_{x=e}$ पर होगा :

- (A) e (B) 2e (C) 4e (D) 8e

Ans. C

Sol. $D^*f(x) = 2f(x) \cdot f'(x)$

$$D^*(x \ln x) = 2x \ln x (1 + \ln x)$$

16. Let $P(x)$ be the polynomial $x^3 + ax^2 + bx + c$, where $a, b, c \in \mathbb{R}$. If $P(-3) = P(+2) = 0$ and $P'(-3) < 0$, which of the following is a possible value of 'c'?

माना $P(x) = x^3 + ax^2 + bx + c$, जहाँ $a, b, c \in \mathbb{R}$. यदि $P(-3) = P(+2) = 0$ और $P'(-3) < 0$, तब 'c' के सम्भव मान है :

- (A) -27 (B) -18 (C) -6 (D) -3

Ans. A

Sol. $P(x) = x^3 + ax^2 + bx + c$

$$P'(x) = 3x^2 + 2ax + b$$

$$P(-3) = 0 \Rightarrow -27 + 9a - 3b + c = 0 \quad \dots(1)$$

$$P(2) = 0 \Rightarrow 8 + 4a + 2b + c = 0 \quad \dots(2)$$

hence (1) - (2) gives

$$5a - 5b - 35 = 0$$

$$a - b = 7 \quad \dots(3)$$

$$P'(-3) = 27 - 6a + b < 0 \Rightarrow 27 - 6(a - b) - 5b < 0 \Rightarrow 27 - 42 - 5b < 0$$

$$\Rightarrow -15 - 5b < 0 \Rightarrow 3 + b > 0 \Rightarrow b > -3, \text{ hence } a - 7 = b > -3$$

$$a - 7 > -3 \quad [\text{from (3)}]$$

$$a > 4 \quad \dots(4)$$

$\therefore$ from (2)

$$8 + 16 - 6 + c_{\max} = 0$$

$$c_{\max} = -18 \Rightarrow c < -18$$

Aliter: $P(x) \equiv (x+3)(x-2)(x-\alpha)$

where α is the third root i.e. $P(x) = x^3 + (1-\alpha)x^2 + (-6-\alpha)x + 6\alpha$ now proceed.

17. f is a differentiable function such that $x = f(t^2)$, $y = f(t^3)$ and $f'(1) \neq 0$ if $\left(\frac{d^2y}{dx^2}\right)_{t=1} =$

f एक अवकलनीय फलन इस प्रकार है कि $x = f(t^2)$, $y = f(t^3)$ तथा $f'(1) \neq 0$ यदि $\left(\frac{d^2y}{dx^2}\right)_{t=1} =$

$$(A) \frac{3}{4} \left(\frac{f''(1) + f'(1)}{(f'(1))^2} \right)$$

$$(B) \frac{3}{4} \left(\frac{f'(1) \cdot f''(1) - f''(1)}{(f'(1))^2} \right)$$

$$(C) \frac{4}{3} \frac{f''(1)}{(f'(1))^2}$$

$$(D) \frac{4}{3} \left(\frac{f'(1)f''(1) - f''(1)}{(f'(1))^2} \right)$$

Ans. A

Sol. $x = f(t^2), \quad y = f(t^3)$

$$\frac{dx}{dt} = 2t f'(t^2) \quad \frac{dy}{dt} = 3t^2 f'(t^3)$$

$$\frac{dy}{dx} = \frac{3t^2 f'(t^3)}{2t f'(t^2)}$$

$$= \frac{\left[3 \frac{dt}{dx} f'(t^3) + 3t f''(t^3) (3t^2) \frac{dt}{dx} \right] 2f'(t^2) - 3t f'(t^3) \left[2f''(t^2) \times 2t \frac{dt}{dx} \right]}{\left[2f'(t^2) \right]^2}$$

$$\Rightarrow \frac{3 \frac{dt}{dx} \left[2f'(t^2) f'(t^3) + 3t^3 f''(t^3) \times 2f'(t^2) - 4t^2 f'(t^3) f''(t^2) \right]}{\left[2f'(t^2) \right]^2}$$

$$= \frac{3}{2f'(1)} \frac{\left[2(f'(1)) \right]^2 + 6f'(1)f''(1) - 4f'(1)f''(1)}{4[f'(1)]^2}$$

$$\Rightarrow \frac{3}{8f'(1)} \left[\frac{\left[2f'(1) \right]^2 + 2f'(1)f''(1)}{f'(1)^2} \right]$$

$$\Rightarrow \frac{3}{4} \left[\frac{f'(1) + f''(1)}{[f'(1)]^2} \right]$$

18. If $P(x)$ is a polynomial of degree less than or equal to 2 and S is the set of all such polynomials so that $P(1) = 1$, $P(0) = 0$ and $P'(x) > 0 \quad \forall x \in [0, 1]$, then :

यदि $P(x)$ एक दो या दो से छोटी घात का बहुपद है एवं S उन सभी बहुपदों के समुच्चय है जिनके लिये $P(1) = 1, P(0) = 0$ तथा $P'(x) > 0 \quad \forall x \in [0, 1]$ हो तो :

(A) $\Sigma = \phi$

(B) $S = \{(1-a)x^2 + ax, 0 < a < 2\}$

(C) $S = \{(1-a)x^2 + ax, a \in (0, \infty)\}$

(D) $S = \{(1-a)x^2 + ax, 0 < a < 1\}$

Ans. B

Sol. Let $P(x) = bx^2 + ax + c$ $b \neq 0$
 $P(0) = 0 \Rightarrow c = 0$
 $P(1) = 1 \Rightarrow a + b = 1$
 $\Rightarrow P(x) = (1-a)x^2 + ax$
 $\Rightarrow P'(x) = 2(1-a)x + a$
 Now $P'(x) > 0$ for $x \in [0, 1]$
 $P'(0) > 0$ & $P'(1) > 0$
 $\Rightarrow a > 0$ & $2(1-a) + a > 0$
 $\Rightarrow 0 < a < 2$
 $\therefore S = \{(1-a)x^2 + ax; 0 < a < 2\}$

19. If $f(x)$ is a continuous and differentiable function and $f(1/n) = 0, \forall n \geq 1$ and $n \in I$, then :

(A) $f(x) = 0, x \in (0, 1]$

(B) $f(0) = 0, f'(0) = 0$

(C) $f'(x) = 0 = f''(x), x \in (0, 1]$

(D) $f(0) = 0$ and $f'(0)$ need not to be zero

यदि $f(x)$ एक सतत् एवं अवकलनीय फलन हो एवं $f(1/n) = 0, \forall n \geq 1$ तथा $n \in I$ तो :

(A) $f(x) = 0, x \in (0, 1]$

(B) $f(0) = 0, f'(0) = 0$

(C) $f'(x) = 0 = f''(x), x \in (0, 1]$

(D) $f'(0) > 0$ तथा $f'(0)$ शून्य होना आवश्यक नहीं है

Ans. B

Sol. $f\left(\frac{1}{n}\right) = 0 \quad \forall n \geq 1, n \in \text{integer}$

$$f\left(\frac{1}{1}\right) = f\left(\frac{1}{2}\right) = f\left(\frac{1}{3}\right) = f\left(\frac{1}{4}\right) = \dots = 0$$

$$\lim_{n \rightarrow \infty} f\left(\frac{1}{n}\right) = 0$$

$$f(0) = 0$$

$$f'(0) = \lim_{n \rightarrow 0} \frac{f(n) - f(0)}{n}$$

$$= \lim_{\frac{1}{n} \rightarrow \infty} \frac{f\left(\frac{1}{n}\right) - 0}{\frac{1}{n}}$$

$$f'(0) = 0$$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

20. For the function $y = f(x) = (x^2 + bx + c)e^x$, which of the following holds?

- (A) if $f(x) > 0$ for all real $x \Rightarrow f'(x) > 0$ (B) if $f(x) > 0$ for all real $x \Rightarrow f'(x) > 0$
 (C) if $f'(x) > 0$ for all real $x \Rightarrow f(x) > 0$ (D) if $f'(x) > 0$ for all real $x \Rightarrow f(x) > 0$

फलन $y = f(x) = (x^2 + bx + c)e^x$, के लिए निम्न में से सत्य है?

- (A) यदि $f(x) > 0$ for all real $x \Rightarrow f'(x) > 0$ (B) यदि $f(x) > 0$ for all real $x \Rightarrow f'(x) > 0$
 (C) यदि $f'(x) > 0$ for all real $x \Rightarrow f(x) > 0$ (D) यदि $f'(x) > 0$ for all real $x \Rightarrow f(x) > 0$

Ans. AC

Sol. $f(x) = (x^2 + bx + c)e^x$

$$\therefore f'(x) = (x^2 + (b+2)x + (b+c))e^x$$

$$f(x) > 0 \quad \text{iff } D = b^2 - 4c < 0$$

$$\text{now } f'(x) > 0 \quad \text{iff } D' = (b+2)^2 - 4(b+c) = D + 4 < 0$$

$$\text{Thus for } f'(x) > 0 \quad D + 4 < 0 \text{ holds.} \quad \Rightarrow \quad D < 0 \Rightarrow f(x) > 0$$

Note that the converse need not be true, e.g. $b = c = 1$, $f(x) > 0$ but $f'(-1) = 0$

21. If $y = \tan x \tan 2x \tan 3x$ then $\frac{dy}{dx}$ has the value equal to :

यदि $y = \tan x \tan 2x \tan 3x$ तब $\frac{dy}{dx}$ बराबर :

- (A) $3 \sec^2 3x \tan x \tan 2x + \sec^2 x \tan 2x \tan 3x + 2 \sec^2 2x \tan 3x \tan x$
 (B) $2y(\operatorname{cosec} 2x + 2 \operatorname{cosec} 4x + 3 \operatorname{cosec} 6x)$
 (C) $3 \sec^2 3x - 2 \sec^2 2x - \sec^2 x$
 (D) $\sec^2 x + 2 \sec^2 2x + 3 \sec^2 3x$

Ans. ABC

Sol. $y = \tan x \tan 2x \tan 3x$

$$\frac{dy}{dx} = \sec^2 x \tan 2x \tan 3x + 2 \tan x \sec^2 2x \tan 3x + 3 \tan x \tan 2x \sec^2 3x$$

$$\frac{dy}{dx} = y \left(\frac{\sec^2 x}{\tan x} + \frac{2 \sec^2 2x}{\tan 2x} + \frac{3 \sec^2 3x}{\tan 3x} \right)$$

$$= y \left(\frac{2}{\sin 2x} + \frac{2 \times 2}{\sin 4x} + \frac{2 \times 3}{\sin 6x} \right)$$

$$= 2y (\operatorname{cosec} 2x + 2 \operatorname{cosec} 4x + 3 \operatorname{cosec} 6x)$$

$$y = \tan x \tan 2x \tan 3x$$

$$\therefore \tan x \tan 2x \tan 3x = \tan 3x - \tan 2x - \tan x$$

$$y = \tan 3x - \tan 2x - \tan x$$

$$\frac{dy}{dx} = 3 \sec^2 3x - 2 \sec^2 2x - \sec^2 x$$

22. Let $f(x) = (x^2 - 3x + 2)(x^2 + 3x + 2)$ and α, β, γ satisfy $\alpha < \beta < \gamma$ are the roots of $f'(x) = 0$ then which of the following is/are correct ([.] denotes greatest integer function)?

माना $f(x) = (x^2 - 3x + 2)(x^2 + 3x + 2)$ तथा $f'(x) = 0$ के मूल α, β, γ हैं, जो कि $\alpha < \beta < \gamma$ को सन्तुष्ट करता है तो निम्न में से सही को चुनिए। ([.] महत्तम पूर्णांक फलन को दर्शाता है)

- (A) $[\alpha] = -2$ (B) $[\beta] = -1$ (C) $[\beta] = 0$ (D) $[\alpha] = 1$

Ans. AC

Sol. $f(x) = (x^2 - 3x + 2)(x^2 + 3x + 2)$

$$= (x^2 + 2 - 3x)(x^2 + 2 + 3x)$$

$$= (x^2 + 2)^2 - 9x^2$$

$$= x^4 + 4 + 4x^2 - 9x^2$$

$$= x^4 - 5x^2 + 4$$

$$f'(x) = 4x^3 - 10x = 0$$

$$2x(2x^2 - 5) = 0$$

$$x = 0, \pm \sqrt{\frac{5}{2}}$$

$$\alpha = -\sqrt{\frac{5}{2}}, \quad \beta = 0, \quad \gamma = \sqrt{\frac{5}{2}}$$

$$[\alpha] = -2 \quad [\beta] = 0 \quad [\gamma] = 1$$

23. If $y = e^{x \sin(x^3)} + (\tan x)^x$ then $\frac{dy}{dx}$ may be equal to :

यदि $y = e^{x \sin(x^3)} + (\tan x)^x$ है तो $\frac{dy}{dx}$ का मान हो सकता है :

(A) $e^{x \sin(x^3)} [3x^3 \cos(x^3) + \sin(x^3)] + (\tan x)^x [\ln \tan x + 2x \operatorname{cosec} 2x]$

(B) $e^{x \sin(x^3)} [x^3 \cos(x^3) + \sin(x^3)] + (\tan x)^x [\ln \tan x + 2x \operatorname{cosec} 2x]$

(C) $e^{x \sin(x^3)} [x^3 \sin(x^3) + \cos(x^3)] + (\tan x)^x [\ln \tan x + 2 \operatorname{cosec} 2x]$

(D) $e^{x \sin(x^3)} [3x^3 \cos(x^3) + \sin(x^3)] + (\tan x)^x \left[\ln \tan x + \frac{x \sec^2 x}{\tan x} \right]$

Ans. AD

Sol. $y = \underbrace{e^{x \sin x^3}}_{y_1} + \underbrace{(\tan x)^x}_{y_2}$

$$y_1 = e^{x \sin x^3}$$

$$\log y_1 = x \sin x^3$$

$$\frac{1}{y_1} \frac{dy_1}{dx} = \sin x^3 + 3x^3 \cos x^3$$

$$\frac{dy_1}{dx} = e^{x \sin x^3} [\sin x^3 + 3x^3 \cos x^3]$$

$$y_2 = (\tan x)^x$$

$$\log y_2 = x \log \tan x$$

$$\frac{1}{y_2} \frac{dy_2}{dx} = \log(\tan x) + \frac{x}{\tan x} \times \sec^2 x$$

$$\frac{dy_2}{dx} = (\tan x)^x [\log(\tan x) + 2x \operatorname{cosec} 2x]$$

$$\frac{dy}{dx} = e^{x \sin(x^3)} [\sin x^3 + 3x^3 \cos x^3] + (\tan x)^x [\log(\tan x) + 2x \operatorname{cosec} 2x]$$

24. Let $f(x) = x + (1-x)x^2 + (1-x)(1-x^2)x^3 + \dots + (1-x)(1-x^2)\dots(1-x^{n-1})x^n$; ($n \geq 4$) then :

माना $f(x) = x + (1-x)x^2 + (1-x)(1-x^2)x^3 + \dots + (1-x)(1-x^2)\dots(1-x^{n-1})x^n$; ($n \geq 4$) है तो :

(A) $f(x) = -\prod_{r=1}^n (1-x^r)$

(B) $f(x) = 1 - \prod_{r=1}^n (1-x^r)$

(C) $f'(x) = (1-f(x)) \left(\sum_{r=1}^n \frac{rx^{r-1}}{(1-x^r)} \right)$

(D) $f'(x) = f(x) \left(\sum_{r=1}^n \frac{rx^{r-1}}{(1-x^r)} \right)$

Ans. BC

Sol. $f(x) = x + (1-x)x^2 + (1-x)(1-x^2)x^3 + \dots + (1-x)(1-x^2)\dots(1-x^{n-1})x^n$

$$1 - f(x) = 1 - x - (1-x)x^2 - (1-x)(1-x^2)x^3 - \dots - (1-x)(1-x^2)\dots(1-x^{n-1})x^n$$

$$1 - f(x) = (1-x) [(1-x^2) - (1-x^2)x^3 - \dots - (1-x^2)(1-x^3)\dots(1-x^{n-1})x^2]$$

$$1 - f(x) = (1-x)(1-x^2) [(1-x^3) - \dots - (1-x^3)\dots(1-x^{n-1})x^n]$$

$$1 - f(x) = (1-x)(1-x^2)(1-x^3)\dots(1-x^n)$$

$$f(x) = 1 - \prod_{r=1}^n (1-x^r)$$

$$\log(1-f(x)) = \log(1-x) + \log(1-x^2) + \log(1-x^3) + \dots + \log(1-x^n)$$

$$\frac{1}{1-f(x)} [-f'(x)] = \frac{-1}{1-x} + \frac{-2x}{1-x^2} + \frac{-3x^2}{1-x^3} + \dots + \frac{-nx^{n-1}}{1-x^n}$$

$$f'(x) = [1-f(x)] \left[\sum_{r=1}^n \frac{rx^{r-1}}{1-x^r} \right]$$